PHYSICS

1)

Find the value of sin 15° = ?

(1) $\frac{1}{2\sqrt{2}}\left(\sqrt{3}-1\right)$

(2) $\frac{1}{\sqrt{2}}\left(\sqrt{3}-1\right)$

(3) $\sqrt{3}-1$

(4) $\frac{1}{2\sqrt{2}}\left(\sqrt{3}+1\right)$

2) Value of sin 4° is approximately equal to :-

(1) 4

(2) $\frac{\pi}{45}$

(3) $\frac{45}{\pi}$

(4) 1

3) If $x_1 = 8\sin\theta$ and $x_2 = 3\cos\theta$ then x_1x_2 = ... :-

(1) $24\sin\theta$

(2) $12\cos2\theta$

(3) $24\sin2\theta$

(4) $12\sin2\theta$

4) The slope of straight line $3y = \sqrt{3}x + 3$ is :-

(1) tan 30°

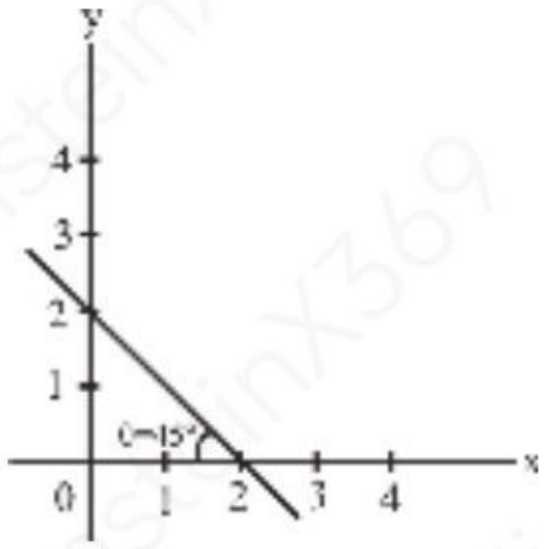
(2) tan 60°

(3) tan 45°

(4) tan 90°

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5) Which equation will represent to the graph



(1) y = x – 2

(2) y = x + 2

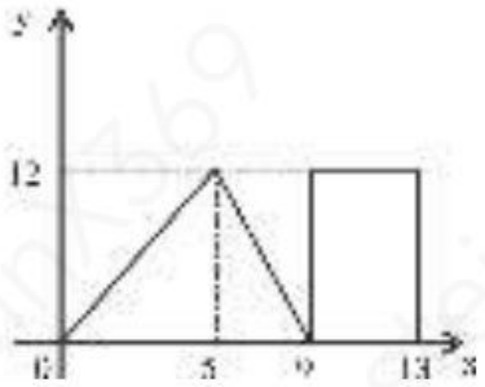
(3) x = y + 2

(4) x = –y + 2

6)

$$I = \int_{9}^{13} y.dx$$

Find , for given x – y curve



(1) 48

(2) 78

(3) 30

(4) None

7) If $y = 3\sin\theta - 4\sin^3\theta$ then find out $\frac{dy}{d\theta}$ at $\theta = \frac{\pi}{3}$

(1) $\frac{3\sqrt{3}}{2}$

(2) – 3

(3) $-\frac{3\sqrt{3}}{2}$

(4) $\frac{1}{2}$

8) If $y = \log_e x + \sin x + e^x$ then $\frac{dy}{dx}$ is

(1) $\frac{1}{x} + \sin x + e^x$

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(2) $\frac{1}{x} - \cos x + e^x$

(3) $\frac{1}{x} + \cos x + e^x$

(4) $\frac{1}{x} - \sin x$

9) Find the value of $\int (3x^2 + 2x + 1)\,dx$

(1) $3x^3 + 2x^2 + x + c$

(2) $6x^2 + 4x^2 + x + c$

(3) $x^3 + x^2 + x + c$

(4) $3x^3 + 2x^2 + c$

10) $1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \ldots\ldots \infty =$

(1) $\frac{4}{3}$

(2) $\frac{3}{2}$

(3) $\frac{5}{2}$

(4) $\frac{13}{9}$

11) Consider a vector $\vec{F}_1 = 2\hat{i} + 5\hat{k}$ and $\vec{F}_2 = 3\hat{j} + 4\hat{k}$. The magnitude of the scalar product of these vectors is :-

(1) 20

(2) 23

(3) $5\sqrt{33}$

(4) 26

12) If A = 1, B = 2 and $\vec{A}.\vec{B} = \sqrt{3}$, then angle between $\vec{A}$ and $\vec{B}$ is :-

(1) 30°

(2) 45°

(3) 60°

(4) 90°

13) If $\hat{a}$ and $\hat{b}$ are non-collinear unit vectors and $\left|\hat{a} + \hat{b}\right| = \sqrt{3}$, then the value of $(\hat{a} - 5\hat{b}).(3\hat{a} + \hat{b})$ is:-

(1) $\frac{41}{2}$

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(2) $\frac{11}{2}$

(3) - 9

(4) $-\frac{41}{2}$

14) If a vector $2\hat{i}+3\hat{j}+8\hat{k}$ is perpendicular to the vector $-4\hat{i}+4\hat{j}+\alpha\hat{k}$. Then the value of α is :

(1) -1

(2) $-\frac{1}{2}$

(3) $\frac{1}{2}$

(4) 1

15) The vector $\vec{B}=5\hat{i}+2\hat{j}-S\hat{k}$ is perpendicular to the vector $\vec{A}=3\hat{i}+\hat{j}+2\hat{k}$ if S =

(1) 1

(2) 4.7

(3) 6.3

(4) 8.5

16) If $\left|\vec{A}\times\vec{B}\right|=\frac{AB}{2}$, then angle between $\vec{A}$ and $\vec{B}$ is :-

(1) 0°

(2) 30°

(3) 45°

(4) 60°

17) The value of $(\vec{A}+\vec{B})\times(\vec{A}-\vec{B})$ is :-

(1) 0

(2) $A^2 - B^2$

(3) $\vec{B}\times\vec{A}$

(4) $2(\vec{B}\times\vec{A})$

18) If $\vec{A}\times\vec{B}=\vec{C}$, then which of the following statements is wrong

(1) $\vec{C}\perp\vec{A}$

(2) $\vec{C}\perp\vec{B}$

(3) $\vec{C}\perp\left(\vec{A}+\vec{B}\right)$

(4) $\vec{C}\perp\left(\vec{A}\times\vec{B}\right)$

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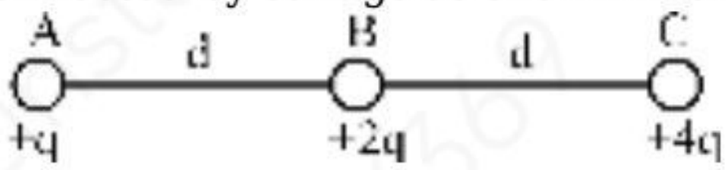
19) If A = 5 units, B = 6 units and $|\vec{A} \times \vec{B}| = 15$ units, then what is the angle between $\vec{A}$ and $\vec{B}$

(1) 30°
(2) 60°
(3) 90°
(4) 120°

20) The area of the parallelogram represented by the vectors $\vec{A} = 2\hat{i} + 3\hat{j}$ and $\vec{B} = \hat{i} + 4\hat{j}$ is

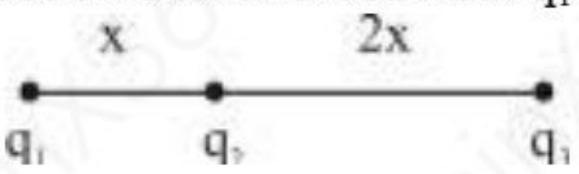
(1) 14 units
(2) 7.5 units
(3) 10 units
(4) 5 units

21) Three charges +q, +2q and +4q are connected by strings as shown in the figure. What is the ratio of tensions in the strings AB and BC.



(1) 1 : 2
(2) 1 : 3
(3) 4 : 1
(4) 4 : 3

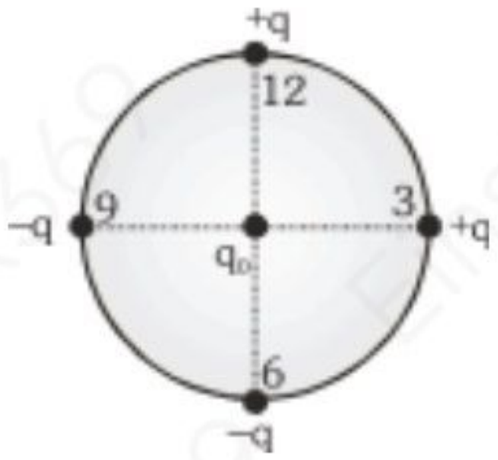
22) Three charged particles are placed on a straight line as shown in the figure. q_1 and q_2 are fixed but q_3 can move. Under the action of the forces from q_1 and q_2 , q_3 is in equilibrium. What is the relation between q_1 and q_2 ?



(1) $q_1 = -9q_2$
(2) $q_1 = \frac{4}{9}q_2$
(3) $q_1 = -\frac{9}{4}q_2$
(4) $q_1 = \frac{9}{4}q_2$

23) Four charges are placed at the circumference of dial clock as shown in figure. If the clock has only hour hand, then the resultant force on a charge q_0 placed at the centre, points in the direction which show the time as :–

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(1) 1:30

(2) 7:30

(3) 4:30

(4) 10:30

24) Particle B of charge Q and mass m is in equilibrium under weight and electrostatics force applied by a fixed charged A, which is directly beneath the particle B as shown in figure. When particle B is disturbed along vertical, the equilibrium is



(1) stable

(2) unstable

(3) neutral

(4) there can not be in equilibrium

25) There are 2 neutral bodies A & B. Both are rubbed with each other so that A acquires positive charge and B acquires negative charge, then body having high work function is :-

(1) A

(2) B

(3) Both have same work function

(4) Can't say

26) Force between A and B is F. If 75% charge of A is transferred to B then force between A and B is:



(1) $\frac{F}{4}$

(2) 4F

(3) F

(4) None

27) Two small conducting spheres of equal radius have charges +10μC and –20μC respectively &

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when placed at a distance R from each other experience force $\vec{F}_1$. If they are brought in contact and separated to the same distance, they experience force $\vec{F}_2$. The ratio of $|\vec{F}_1|$ to $|\vec{F}_2|$ is :-

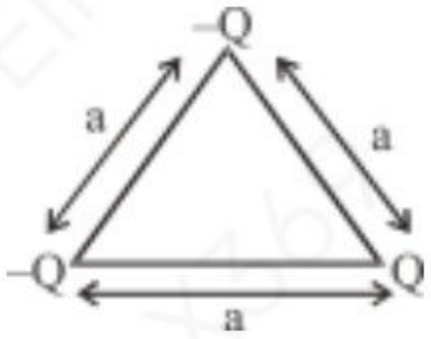
(1) 1 : 8

(2) 8 : 1

(3) 1 : 2

(4) 2 : 1

28) Find magnitude of net force on charge Q :



(1) $\frac{KQ^2}{a^2}$

(2) $\frac{2KQ^2}{a^2}$

(3) $\frac{\sqrt{3}KQ^2}{a^2}$

(4) $\frac{3KQ^2}{a^2}$

29) Four charges each of 2μC are placed at x = 0,2,4,8 cm on x-axis. The force exerted on the charge placed at x = 2 cm will be :-

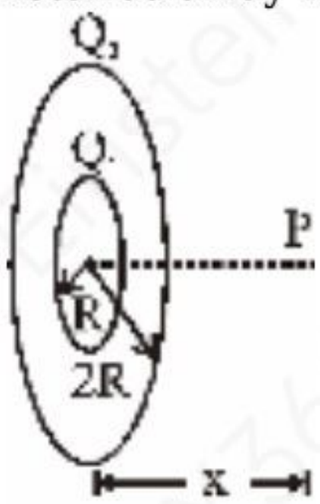
(1) zero

(2) 10 N

(3) 5 N

(4) 10^{-3} N

30) Two uniformly charged rings of radius R and 2R are placed co-axially as shown in figure. If net electric field at point P, which is x (x < < R) distance away from common centre 'O' is zero then



which relation is correct for charge in rings :

(1) $Q_1 = -8Q_2$

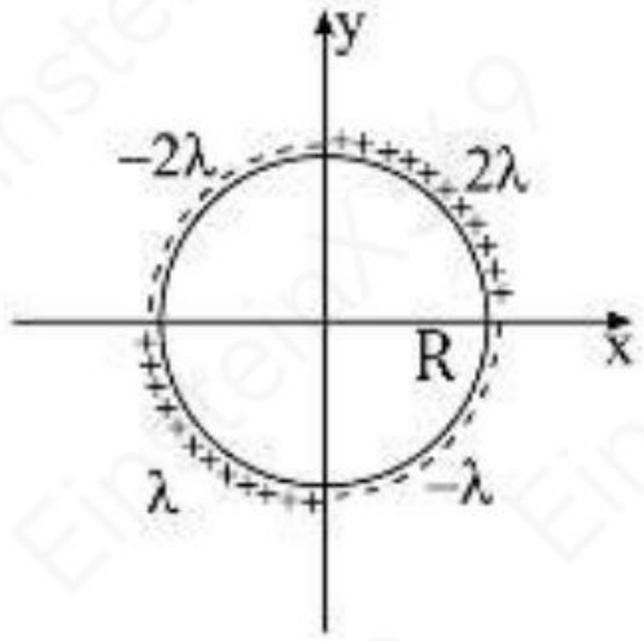
(2) $Q_2 = -8Q_1$

(3) $Q_1 = -4Q_2$

(4) $Q_2 = -4Q_1$

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31) The charge per unit length of the four quadrant of the ring is 2λ, – 2λ, λ and – λ respectively. The



magnitude of electric field at the centre is :

(1) $\frac{2K\lambda}{r}$

(2) $\frac{K\lambda}{r}$

(3) $\frac{4K\lambda}{r}$

(4) zero

32) A rod of length L has a total charge Q distributed uniformly along its length. It is bent in the shape of a semicircle. Find the magnitude of the electric field at the centre of curvature of the semicircle.

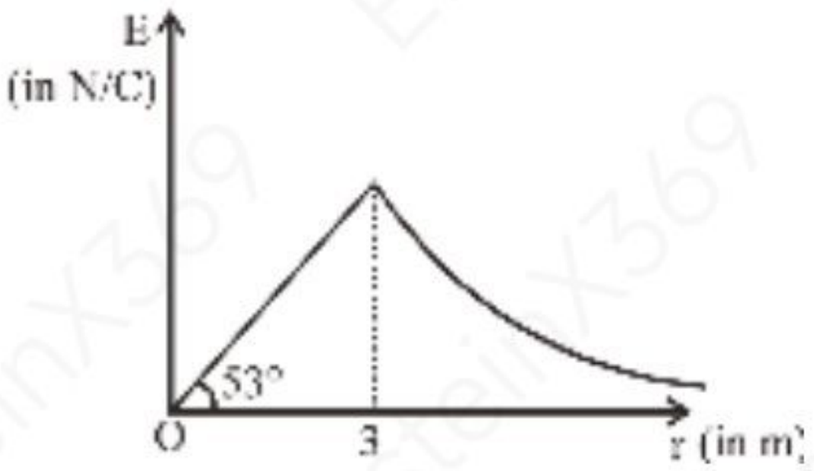
(1) $Q/2\epsilon_0 L^2$

(2) $Q/\epsilon_0 L^2$

(3) $\frac{2Q}{\epsilon_0 L^2}$

(4) Zero

33) We have a solid non conducting uniformly charged sphere. Its electric field variation with distance from centre is shown in graph then the net charge inside the sphere (in S.I. units) is :-



(1) $144\ \pi\epsilon_0$

(2) $100\ \pi\epsilon_0$

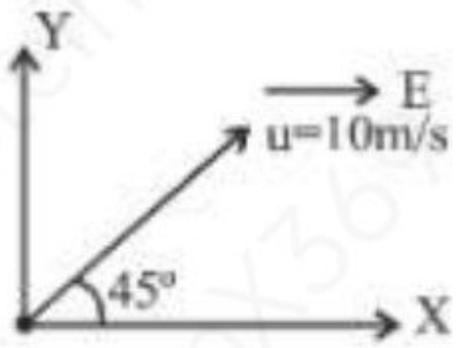
(3) $32\ \pi\epsilon_0$

(4) $48\ \pi\epsilon_0$

34) A charged particle of mass m = 2 kg and charge 1 μC is thrown from a horizontal ground at an angle of 45º with horizontal with a speed of 10 m/s. In space a horizontal electric field E = 2 ×

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10^7 N/C exists. Find the range of the projectile.



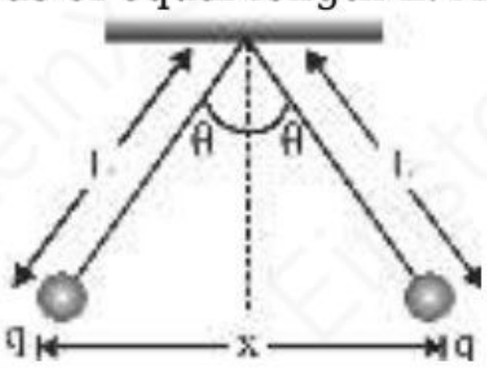
(1) 5 m

(2) 30 m

(3) 10 m

(4) 20 m

35) In the given figure two tiny conducting balls of identical mass m and identical charge q hang from non-conducting threads of equal length L. Assume that θ is so small that tan θ ≈ sinθ, then for equilibrium x is equal to :-



(1) $\left(\frac{q^2L}{2\pi\varepsilon_0 mg}\right)^{\frac{1}{3}}$

(2) $\left(\frac{qL^2}{2\pi\varepsilon_0 mg}\right)^{\frac{1}{3}}$

(3) $\left(\frac{q^2L^2}{4\pi\varepsilon_0 mg}\right)^{\frac{1}{3}}$

(4) $\left(\frac{q^2L}{4\pi\varepsilon_0 mg}\right)^{\frac{1}{3}}$

36) If the volume charge density is doubled, the electric field at a fixed point inside the sphere:

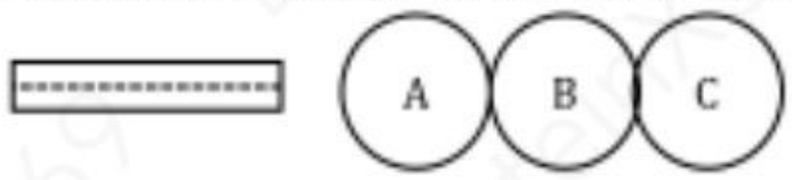
(1) Remains same

(2) Is halved

(3) Is doubled

(4) Is quadrupled

37) Find the charge on the three spheres ABC respectively as shown figure due to induction



Arjun that all 3 sphere are conductor

(1) +ve, 0, –ve

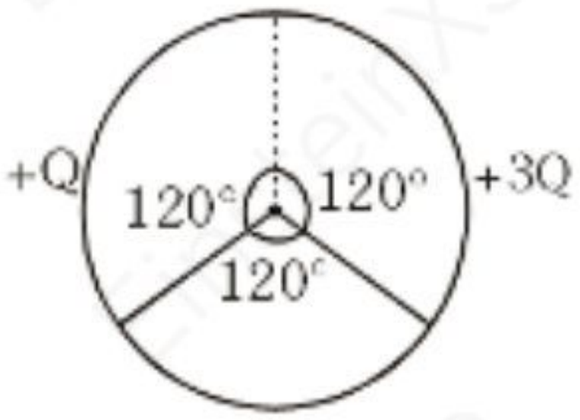
(2) –ve, 0, –ve

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(3) +ve, +ve, –ve

(4) +ve, –ve, –vc

38) Figure shows three circular arcs, each of radius R and total charge as indicated. The net electric potential at the centre of curvature is



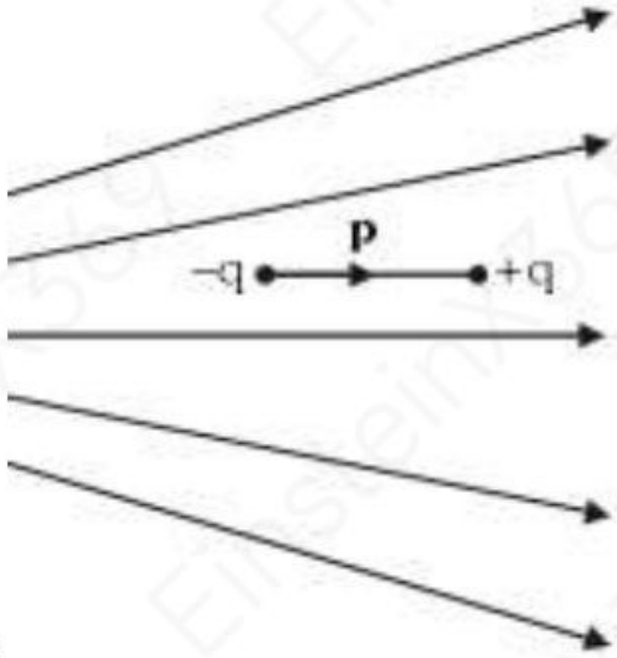
(1) $\frac{Q}{4\pi\in_0 R}$

(2) $\frac{Q}{2\pi\in_0 R}$

(3) $\frac{2Q}{\pi\in_0 R}$

(4) $\frac{Q}{\pi\in_0 R}$

39) Figure shows electric field lines in which an electric dipole **p** is placed as shown. Which of the following statements is correct?



(1) The dipole will not experience any force.

(2) The dipole will experience a force towards right.

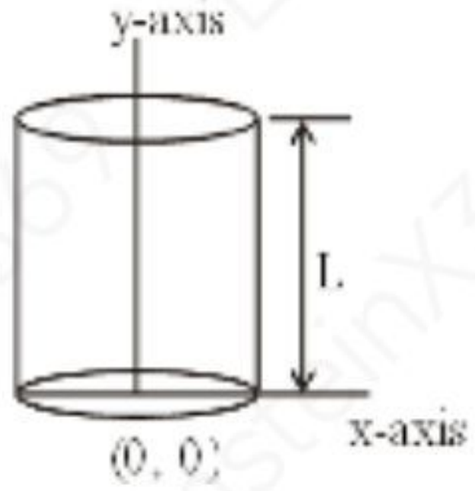
(3) The dipole will experience a force towards left.

(4) The dipole will experience a force upwards.

40) A long charged wire is kept on +y axis, with one end at origin. Its linear charge density is given

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0



as $\lambda = \lambda y$. Calculate electric flux passing through the cylinder.

(1) $\frac{\lambda_0 L^2}{\epsilon_0}$

(2) $\frac{\lambda_0 L^2}{2\epsilon_0}$

(3) $\frac{2\lambda_0 L^2}{\epsilon_0}$

(4) None

41) A and B are two points on the axis and the perpendicular bisector respectively of an electric dipole. A and B are far away from the dipole and at equal distance from it. The field at A and B are $\vec{E}_A$ and $\vec{E}_B$.

(1) $\vec{E}_A = \vec{E}_B$

(2) $\vec{E}_A = 2\vec{E}_B$

(3) $E_A = -2E_B$

(4) $|E_B| = \frac{1}{2}|E_A|$, and $\vec{E}_B$ is perpendicular to $\vec{E}_A$

42) In the following figure two parallel metallic plates are maintained at different potential. If an electron is released midway between the plates, it will move : -



(1) Right ward at constant speed

(2) Left ward at constant speed

(3) Accelerated right ward

(4) Accelerated left ward

43) Force between two point charges q_1 and q_2 placed in vacuum at 'r' cm apart is F. Force between them when placed in a medium having dielectric K = 5 at 'r/5' cm apart will be : -

(1) F/25

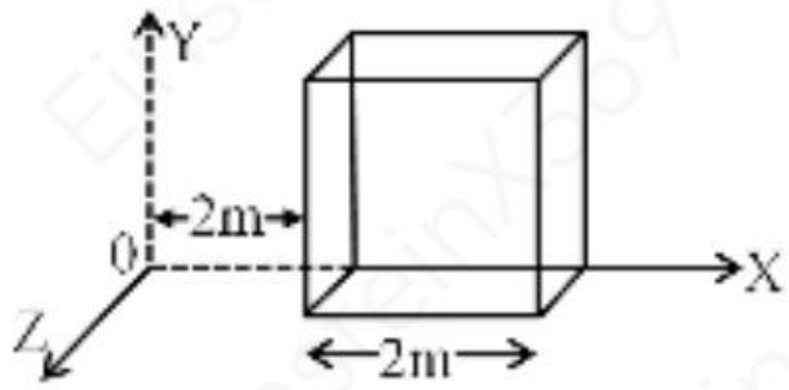
(2) 5F

(3) F/5

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(4) 25F

44) An electric field $\vec{E} = (2x\hat{i})NC^{-1}/C$.



(1) 30

(2) 16

(3) 14

(4) 12

45) Dimension of electric potential : -

(1) $[M^1L^2T^{-2}A^{-1}]$

(2) $[M^1L^2T^{-3}A^{-1}]$

(3) $[M^1L^2T^{-3}A^1]$

(4) $[M^1L^2T^{-1}A^{-1}]$

CHEMISTRY

1) Which statement about molecularity of a reaction is wrong ?

(1) It is the number of molecules of the reactants taking part in a single step of reaction

(2) It is calculated from the reaction mechanism

(3) It may be either whole number or fractional

(4) None of the above

2) Higher order (> 3) reactions are rare due to:

(1) low probability of simultaneous collision of all the reacting species

(2) increase in entropy and activation energy as more molecules are involved

(3) shifting of equilibrium towards reactants due to elastic collisions

(4) loss of active species on collision

3) The rate constant of nth order has unit:

(1) $litre^{1-n}\ mol^{1-n}\ sec^{-1}$

(2) $mol^{n-1}\ litre^{n-1}\ sec^{-1}$

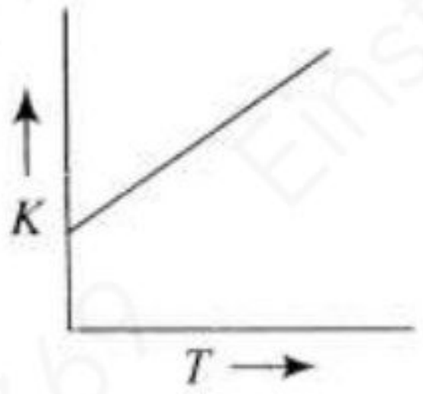
(3) $mol^{1-n}\ litre^{n-1}\ sec^{-1}$

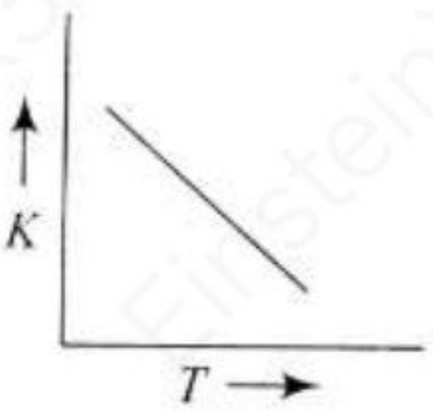
(4) none of these

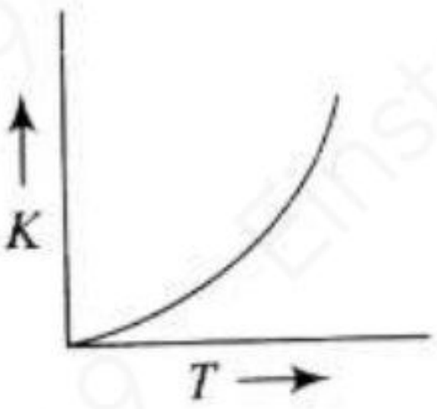
4) Plots showing the variation of the rate constant (k) with temperature (T) are given below. The plot

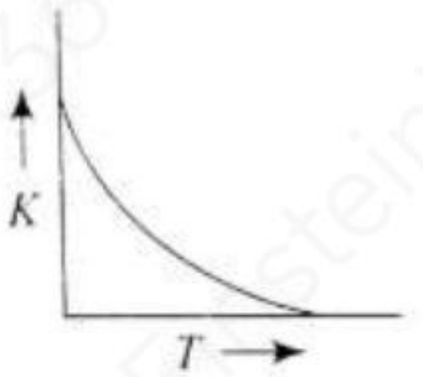
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that follows Arrhenius equation is:

(1) 

(2) 

(3) 

(4) 

5) For producing the effective collisions the colliding molecules must have:

(1) a certain minimum amount of energy
(2) energy lesser than thershold energy
(3) improper orientation
(4) proper orientation and energy equal or greater than threshold energy

6) For the reaction $NO_2 + F_2 \rightarrow 2NO_2F$, following mechanism has been provided,

$$NO_2 + F_2 \xrightarrow{\text{Slow}} NO_2F + F$$

$$NO_2 + F \xrightarrow{\text{Fast}} NO_2F$$

Thus, rate expression of the above reaction can be written as:

(1) $r = k[NO_2]^2 [F_2]$
(2) $r = k [NO_2][F_2]$
(3) $r = k[NO_2]$
(4) $r = k[F_2]$

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7) For a reaction for which the activation energies of forward and reverse reactions are equal ?

(1) $\Delta H = 0$

(2) $\Delta S = 0$

(3) The order is zero

(4) There is no catalyst

8) The differential rate expression for the reaction $H_2 + I_2 \rightarrow 2HI$ is:

(1) $\frac{-d[H_2]}{dt} = \frac{-d[I_2]}{dt} = \frac{-d[HI]}{dt}$

(2) $\frac{d[H_2]}{dt} = \frac{d[I_2]}{dt} = \frac{d[HI]}{dt}$

(3) $\frac{1}{2}\frac{d[H_2]}{dt} = \frac{1}{2}\frac{d[I_2]}{dt} = \frac{d[HI]}{dt}$

(4) $-2\frac{d[H_2]}{dt} = -2\frac{d[I_2]}{dt} = \frac{d[HI]}{dt}$

9) For the reaction $N_2 + 3H_2 \rightarrow 2NH_3$, $\frac{d[NH_3]}{dt} = 2 \times 10^{-4} Ms^{-1}$. Therefore, the rate $-\frac{d[N_2]}{dt}$ is given as:

(1) 10^{-4} M s^{-1}

(2) 10^{4} M s^{-1}

(3) 10^{-2} M s^{-1}

(4) 10^{-4} s M^{-1}

10) For the reaction, $N_2O_5 \rightarrow 2NO_2 + \frac{1}{2}O_2$

Given, $-\frac{d[N_2O_5]}{dt} = k_1[N_2O_5]$

$\frac{d[NO_2]}{dt} = k_2[N_2O_5]$

and $\frac{d[O_2]}{dt} = k_3[N_2O_5]$

The relation in between k_1, k_2 and k_3 is :

(1) $2k_1 = k_2 = 4k_3$

(2) $k_1 = k_2 = k_3$

(3) $2k_1 = 4k_2 = k_3$

(4) none of the above

11) Compounds 'A' and 'B' react according to the following chemical equation.

$A(g) + 2B(g) \rightarrow 2C(g)$

Concentration of either 'A' or 'B' were changed keeping the concentrations of one of the reactants constant and rates were measured as a function of initial concentration. Following results were obtained. Choose the correct option for the rate equations for this reaction.

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Experiment	Initial concentration of [A]/mol L^{-1}	Initial concentration of [B]/mol L^{-1}	Initial rate of formation of [C]/mol $L^{-1}s^{-1}$
1.	0.30	0.30	0.10
2.	0.30	0.60	0.40
3.	0.60	0.30	0.20

(1) Rate = $k[A]^2[B]$

(2) Rate = $k[A][B]^2$

(3) Rate = $k[A][B]$

(4) Rate = $k[A]^2[B]^0$

12) The rate of reaction between two reactants A and B decreases by a factor 4 if the concentration of reactant B is doubled. The order of this reaction with respect to B is :

(1) –1

(2) –2

(3) 2

(4) 1

13) The rate constant k_1 and k_2 for two different reactions are $10^{16}\ e^{-2000/T}$ and $10^{15}\ e^{-1000/T}$, respectively. The temperature at which $k_1 = k_2$ is :

(1) $\frac{2000}{2.303}K$

(2) 2000 K

(3) $\frac{1000}{2.303}K$

(4) 1000 K

14) Half-life period of a first order reaction is 1386 seconds. The specific reaction rate of the reaction is :

(1) $5.0 \times 10^{-2}\ s^{-1}$

(2) $5.0 \times 10^{-3}\ s^{-1}$

(3) $5 \times 10^{-5}\ s^{-1}$

(4) $0.5 \times 10^{-3}\ s^{-1}$

15) In a zero-order reaction for every 10°C rise of temperature, the rate is doubled. If the temperature is increased from 10°C to 100°C, the rate of the reaction will become:

(1) 64 times

(2) 128 times

(3) 256 times

(4) 512 times

16) Activation energy (E_a) and rate constants (k_1 and k_2) of a chemical reaction at two different temperatures (T_1 and T_2) are related by :

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(1) $\ln\frac{k_2}{k_1} = \frac{-E_a}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$

(2) $\ln\frac{k_2}{k_1} = \frac{E_a}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$

(3) $\ln\frac{k_2}{k_1} = \frac{E_a}{R}\left(\frac{1}{T_2} + \frac{1}{T_1}\right)$

(4) $\ln\frac{k_2}{k_1} = \frac{E_a}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$

17) When initial concentration of a reactant is doubled in a reaction, its rate is not affected. The order of the reaction is :

(1) First

(2) Second

(3) More than zero but less than first

(4) Zero

18) Consider the reaction

$$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$$

The equality relationship between $\frac{d[NH_3]}{dt}$ and $-\frac{d[H_2]}{dt}$ is :

(1) $+\frac{d[NH_3]}{dt} = -\frac{3}{2}\frac{d[H_2]}{dt}$

(2) $\frac{d[NH_3]}{dt} = -\frac{d[H_2]}{dt}$

(3) $\frac{d[NH_3]}{dt} = -\frac{1}{3}\frac{d[H_2]}{dt}$

(4) $+\frac{d[NH_3]}{dt} = -\frac{2}{3}\frac{d[H_2]}{dt}$

19) During the kinetic study of the reaction $2A + B \rightarrow C + D$ following results were obtained.

S.N.	[A] in M	[B] in M	Initial rate in M s^{-1}
I	0.1	0.1	6.0×10^{-3}
II	0.3	0.2	7.2×10^{-2}
III	0.3	0.4	2.88×10^{-1}
IV	0.4	0.1	2.40×10^{-2}

On the basis of above data which one is correct ?

(1) $r = k[A]^2[B]$

(2) $r = k[A][B]$

(3) $r = k[A]^2[B]^2$

(4) $r = k[A][B]^2$

20) In question given below a statement **(S)** and explanation **(E)** is given. Choose the correct answers:

S: Every collision between molecules leads to a chemical reaction.

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E: Only those molecules react during collisions which acquire threshold energy level and have proper orientation

(1) **S** is correct but **E** is wrong.

(2) **S** is wrong but **E** is correct

(3) Both **S** and **E** are correct and **E** is correct explanation of **S.**

(4) Both **S** and **E** are correct but **E** is not correct explanation of **S**.

21) In a first order reaction A → B, if k is the rate constant and initial concentration of the reactant is 0.5 M, then half-life is :

(1) $\frac{\ln 2}{k}$

(2) $\frac{\ln 2}{k\sqrt{0.5}}$

(3) $\frac{\log_{10} 2}{k}$

(4) $\frac{0.693}{0.5\,k}$

22) Activation energy of a chemical reaction can be determined by :

(1) evaluating rate constants at two different temperatures

(2) evaluating velocities of reaction at two different temperatures

(3) evaluating rate constant at standard temperature

(4) changing concentration of reactants

23) In the respect of the equation $k = Ae^{-E_a/RT}$ in chemical kinetics, which one of the following statements is correct

(1) k is equilibrium constant

(2) A is adsorption factor

(3) E_a is energy of activation

(4) R is Rydberg constant

24) Normality of 10% (w/v) H_2SO_4 solution is nearly :

(1) 0.1

(2) 0.2

(3) 0.5

(4) 2

25) 1000 g $CaCO_3$ solution contains 10 g carbonate. The concentration of solution is :

(1) 10000 ppm

(2) 1000 ppm

(3) 100 ppm

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(4) 10 ppm

26) On mixing 10 mL of carbon tetrachloride with 10 mL of benzene both being non-polar the total volume of the solution is:

(1) > 20 mL

(2) = 20 mL

(3) < 20 mL

(4) Can't be predicted

27) The value of K_b for water is 1.86, calculated from Glucose solution. The value of K_b for water calculated for NaCl solution will be:

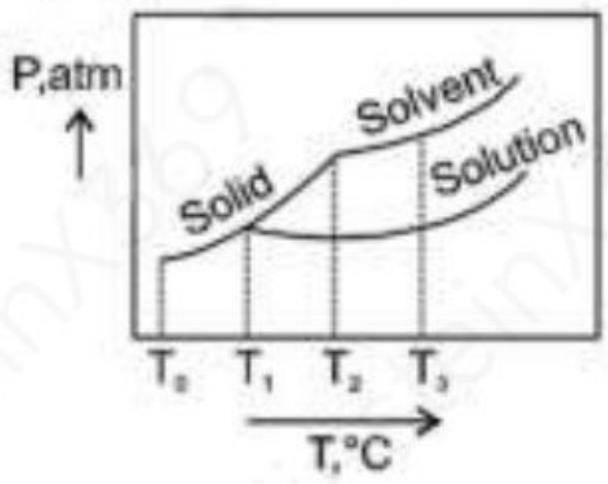
(1) < 1.86

(2) = 1.86

(3) > 1.86

(4) Zero

28) What is the normal freezing point of the solution represented by the phase diagram ?



(1) T_1

(2) T_2

(3) T_3

(4) T_0

29) Mixture of volatile components A and B has total vapour pressure (in torr):
$P = 254 - 119\ X_A$

Where, X_A is mole fraction of A in the mixture. Hence, P^0_A and P^0_B are (in torr)

(1) 254, 135

(2) 119, 254

(3) 154, 119

(4) 135, 254

30) An ideal solution contains two volatile liquids A (P^0 = 100 torr) and B (P^0 = 200 torr). If mixture contain 1 mole of A and 4 mole of B then total vapour pressure of the solution is:

(1) 150 torr

(2) 180 torr

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(3) 188.88 torr

(4) 190 torr

31) For a solution of two liquids A and B showing negative deviation from Raoult's law, which of the following statements are true ?

(1) $\Delta H_{mix} < 0$

(2) $\Delta V_{mix} < 0$

(3) $P < P_A^0 X_A + P_B^0 X_B$

(4) $\Delta S_{mix} < 0$

(1) 1, 2, 4

(2) 2, 3, 4

(3) 1, 2, 3

(4) 1, 3, 4

32) Two liquids having vapour pressures P_1^0 and P_2^0 in pure state in the ratio of 2 : 1 are mixed in the molar ratio of 1 : 2. The ratio of their moles in the vapour state would be:

(1) 1 : 1

(2) 1 : 2

(3) 2 : 1

(4) 3 : 2

33) Match the Column-I with Column-II :

Column-I	Column-II
(A) Molality	(p) Equal to the relative lowering of vapour pressure
(B) Molarity	(q) Varies with temperature
(C) Mole fraction of solute	(r) Elevation in boiling point is proportional to it
(D) Normality	(s) Number of equivalents of solute dissolved per litre of solution

(1) (A) → (s); (B) → (p) ; (C) → (r); (D) → (q)

(2) (A) → (q); (B) → (p) ; (C) → (r); (D) → (s)

(3) (A) → (r); (B) → (q) ; (C) →(p); (D) → (q, s)

(4) (A) → (p); (B) → (q) ; (C) → (s); (D) → (r)

34) Vapour pressure of a solution having solid as solute and liquid as solvent is :

(1) inversely proportional to the mole fraction of the solute

(2) directly proportional to the mole fraction of the solvent

(3) inversely proportional to the mole fraction of the solvent

(4) directly proportional to the mole fraction of solute

35) Calculate the mass of a non-volatile solute (molar mass 40 g mol^{-1}) which should be dissolved in 114 g octane to reduce its vapour pressure to 80%:

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(1) 5 g
(2) 2.5 g
(3) 20 g
(4) 10 g

36) Decrease in the vapour pressure of the solvent depends on:

(1) nature of non-volatile solute
(2) quantity of non-volatile solute
(3) nature of solution
(4) quantity of solution

37) **Assertion:** When a solution is separated from the pure solvent by a semi-permeable membrane, the solvent molecules pass through it from pure solvent side to the solution side.
Reason: Diffusion of solute occurs from a region of high concentration solution to a region of low concentration of solution through semi-permeable membrane.

(1) Assertion and reason both are correct statements and reason is correct explanation for assertion.
(2) Assertion and reason both are correct statements but reason is not correct explanation for assertion.
(3) Assertion is correct statement but reason is wrong statement.
(4) Assertion and reason both are incorrect statements

38) A solution of acetone in ethanol :

(1) obeys Raoult's law
(2) shows a negative deviation from Raoult's law
(3) shows a positive deviation from Raoult's law
(4) behaves like a near ideal solution

39) The mixture of two liquids boils without changing its composition is called:

(1) zeotropic liquid mixture
(2) binary liquid mixture
(3) azeotropic liquid mixture
(4) stable structure complex

40) Calculate elevation in boiling point for 2 molal aqueous solution of glucose:
(Given: $K_b(H_2O) = 0.5\ K\ kgmol^{-1}$).

(1) 1 °C
(2) 4 °C
(3) 3 °C
(4) 2 °C

41) Cryoscopic constant of a liquid is:

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(1) decrease in freezing point when 1 gram of solute is dissolved per kg of the solvent
(2) decrease in the freezing point when 1 mole of solute is dissolved per kg of the solvent
(3) the elevation for 1 molar solution
(4) a factor used for calculation of elevation in boiling point

42) What is ΔT_b for 0.2 molal aqueous solution of urea? (K_b of H_2O = 0.52)

(1) 0.104 K
(2) 101.04 K
(3) 0.506 K
(4) 5.06 K

43) **Assertion(A):** Molarity of a solution changes with temperature.
Reason (R): Molarity is a colligative property.

(1) Both Assertion (A) and Reason (R) are correct statements, and Reason (R) is the correct explanation of the Assertion (A).
(2) Both Assertion (A) and Reason (R) are correct statements, but Reason (R) is not the correct explanation of the Assertion (A).
(3) Assertion (A) is correct, but Reason (R) is wrong statement.
(4) Assertion (A) is wrong, but Reason (R) is correct statement.

44) In which of the following cases blood cells will shrink:

(1) when placed in water containing more than 0.9% (mass/ volume) NaCl solution.
(2) when placed in water containing less than 0.9% (mass /volume) NaCl solution.
(3) when placed in water containing 0.9% (mass/volume) NaCl solution.
(4) when placed in distilled water.

45) Out of the following concentration terms :-
(a) mass% (b) ppm (by mass) (c) mole fraction
(d) molality (e) molarity
Those which are independent of temperature is:-

(1) a, b, d & e
(2) b, e
(3) a, b, c & d
(4) a, c & d only

BIOLOGY

1) Read the following six statements carefully and choose the correct combination :
1. A typical eukaryotic cell divides once in approximately every 24 hours.
2. *Yeast* completes the cell cycle in about 90 minutes.
3. Interphase represents the phase between two successive M phases.
4. In the 24-hour average cell cycle of a human cell, cell division proper lasts for only about an hour.

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5. During the G_1 phase, the cell is metabolically active but does not replicate its DNA.
6. During S phase, if the initial DNA content is 2C, then it becomes 4C, but the chromosome number remains the same.

(1) Only statements 1, 2, 4, and 5 are correct

(2) Only statements 2, 3, 4, and 6 are correct

(3) Only statements 1, 3, 5, and 6 are correct

(4) All six statements are correct

2) **Assertion (A):** DNA replication occurs during interphase.
Reason (R): M phase includes synthesis of DNA.

(1) Both A and R are true and R is the correct explanation of A.

(2) Both A and R are true but R is not the correct explanation of A.

(3) A is true but R is false.

(4) A is false but R is true.

3)

Which of the following statements are true?
1. Karyokinesis results in the separation of sister chromatids.
2. Centriole duplication occurs in the G_2 phase of the cell cycle.
3. The interphase is the longest phase of the cell cycle.
4. Mitosis in plants occurs only in diploid cells.
5. Cytokinesis is the division of cytoplasm.

(1) 1, 3, and 5 only

(2) 1, 2, 4, and 5 only

(3) 2, 3, and 5 only

(4) 1, 3, 4, and 5 only

4) If a student observes a cell with condensed chromosomes arranged at the center of the cell, spindle fibres attached at kinetochores, and no visible nuclear membrane, what phase is the cell in?

(1) Prophase

(2) Metaphase

(3) Anaphase

(4) Telophase

5) Which enzyme is responsible for catalyzing crossing over during meiosis?

(1) DNA polymerase

(2) Ligase

(3) Recombinase

(4) Helicase

6) During metaphase II, chromosomes align at the _____ and microtubules from _____ poles

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attach to kinetochores of sister chromatids. The result of meiosis II is the formation of _____ haploid daughter cells.

(1) Equator; opposite; four

(2) Poles; same; two

(3) Metaphase plate; opposite; two

(4) Equator; same; four

7) In a cell undergoing meiosis II, spindle fibers attach to kinetochores on chromosomes aligned at the equator. If a mutation caused kinetochores to fail to form, which of the following consequences is most likely?

(1) Chromosomes would not condense properly

(2) Sister chromatids would fail to separate during anaphase II

(3) Homologous chromosomes would fail to pair in meiosis II

(4) Cytokinesis would be inhibited

8) Leptotene : Chromosome visibility :: Pachytene : ?

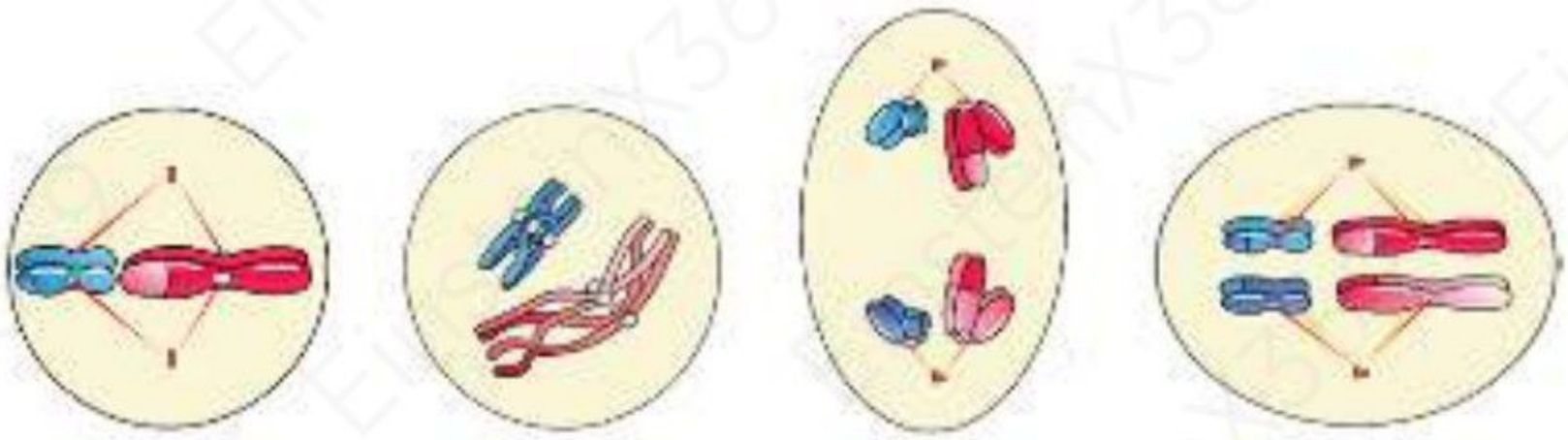
(1) Synapsis initiation

(2) Chiasmata appearance

(3) Crossing over

(4) Nuclear envelope breakdown

9) Arrange the following figures (of stages of meiosis) in the order of their occurrence and choose the correct option.



(1) A → C → D → B

(2) B → D → C → A

(3) B → D → A → C

(4) C → B → D → A

10) Interkinesis is a -

(1) Stage between meiosis I and meiosis II

(2) Stage between two mitotic divisions

(3) Interphase

(4) Both (2) and (3)

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11) During mitosis, nuclear envelope, nucleolus begin to form and ER to reappear at _____.

(1) Prophase
(2) Late metaphase
(3) Anaphase
(4) Telophase

12) Which of the following statement(s) is/are true ?

(1) Cell plate represents the middle lamella between the walls of two adjacent plant cells.
(2) At the time of cytokinesis, organelles like mitochondria and plastids get distributed between the daughter cells.
(3) Cytokinesis in plant cell is centrifugal and takes place by cell-plate formation while animal cells by furrowing/cleavage and is centripetal.
(4) All of the above

13) Chiasmata are first seen in _____.

(1) Leptotene
(2) Diplotene
(3) Pachytene
(4) Zygotene

14) Select the correct statement with respect to mitosis.

(1) Chromatids separate but remain in the centre of the cell in anaphase.
(2) Chromatids start moving towards opposite poles in telophase.
(3) Golgi complex and endoplasmic reticulum are still visible at the end of prophase.
(4) Chromosomes move to the spindle equator and get aligned along equatorial plate in metaphase.

15) Which of the following statements related to G_0 stage of the cell cycle are correct?
(i) It is a quiescent stage.
(ii) In this phase, cell cycle is stopped.
(iii) G_0 cells do not grow or proliferate but metabolically active.
(iv) G_0 cells can divide in response to some stimulus.

(1) (i) and (ii)
(2) (i), (ii) and (iii)
(3) Only (i) and (iv)
(4) All of the above

16) Match the description (given in column I) with correct stage of prophase I (given column II) and choose the correct option.

Column-I		**Column-II**	
A	Chromosomes are moved to spindle equator	I	Pachytene

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B	Centromere splits and chromatids apart	II	Zygotene
C	Pairing between homologous chromosomes takes place	III	Anaphase
D	Crossing between homologous chromosomes	IV	Metaphase

(1) A – I; B – II; C – III; D – IV

(2) A – II; B – III; C – IV; D – I

(3) A – IV; B – III; C – II; D – I

(4) A – III; B – I; C – IV; D – II

17) **Assertion:** Karyokinesis follows cytokinesis.
Reason: Karyokinesis is the division of cytoplasm into two daughter cells.

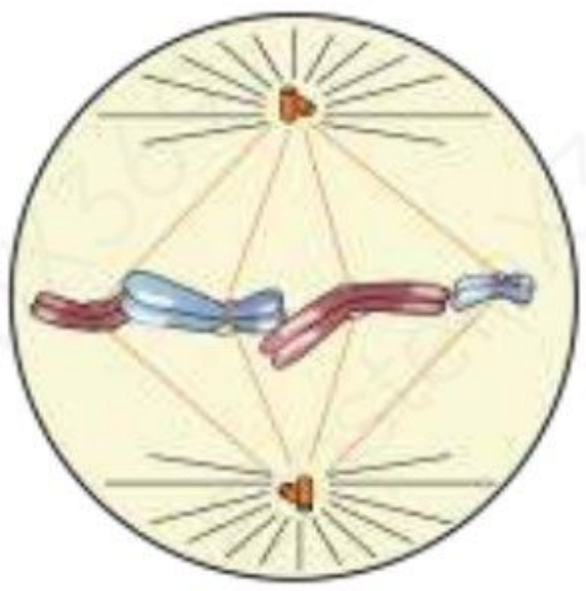
(1) Both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.

(2) Both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

18) A stage of mitosis is shown in the given diagram. Identify stage with its characteristics?



(1) Late prophase – Chromosomes move to spindle equator.

(2) Metaphase – Spindle fibres attached to kinetochores, centromeres split and chromatids separate.

(3) Metaphase – chromosomes moved to spindle equator chromosomes made up of two sister chromatids.

(4) Anaphase – centromeres split and chromatids separate and start moving away.

19) Which of the following is the result when karyokinesis is not followed by cytokinesis ?

(1) Synaptonemal complex

(2) Syncytium

(3) Recombination nodules

(4) Terminalisation

20) In animal cells, cytokinesis involves -

(1) The separation of sister chromatids.

(2) The contraction of the contractile ring of microfilament.

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(3) Depolymerization of kinetochore microtubules.

(4) A protein kinase that phosphorylates other enzymes.

21) In Meiosis -

(1) Division of nucleus twice but DNA replicates only once.

(2) Division of nucleus twice and DNA replicates twice.

(3) Division of nucleus once and replication of DNA is also once.

(4) Division of nucleus once and DNA - replication is twice.

22) The following events occur during the process of meiosis. Arrange the following events of meiosis in the correct sequence.

I. Terminalisation

II. Crossing over

III. Synapsis

IV. Disjunction of genomes

(1) I, II, III, IV

(2) II, III, IV, I

(3) III, II, I, IV

(4) IV, I, II, III

23) In meiosis, division is -

(1) I^{st} reductional and IInd equational

(2) I^{st} equational and IInd reductional

(3) Both reductional

(4) Both equational

24) Which of the following cellular structures always disappears during mitosis and meiosis?

(1) Plastid and nuclear membrane

(2) Nucleolus and nuclear membrane

(3) Endoplasmic reticulum and mitochondria

(4) Endoplasmic reticulum and plasma membrane

25) The synaptonemal complex appears -

(1) Between homologous chromosomes

(2) In zygotene stage

(3) Bivalents are formed

(4) All of the above

26) Choose the correct statement -

(1) Simple squamous epithelium is specialized for secretion.

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(2) Cuboidal epithelium lining PCT mainly facilitates rapid diffusion.

(3) Simple cuboidal epithelium lining the proximal convoluted tubule bears microvilli to increase reabsorption.

(4) Stratified epithelium lines the alveoli of lungs.

27) Adhering junctions : Cementing adjacent cells :: Tight junctions : ?

(1) Absorption

(2) Communication

(3) Preventing leakage between adjacent cells

(4) Secretion

28) Choose the correct statements:
A. Fibroblasts are the principal cells responsible for producing fibres and ground substance in connective tissue.
B. Adipose tissue acts as an energy reserve and provides insulation against heat loss.
C. Tendons connect bone to bone.
D. Dense regular connective tissue contains collagen fibres arranged in parallel bundles.

(1) A, B and D only

(2) A and C only

(3) B, C and D only

(4) A, B, C and D

29) The muscle tissue shown in the diagram is correctly identified as:



(1) Skeletal muscle

(2) Voluntary muscle

(3) Cardiac muscle

(4) Smooth muscle

30) Choose the correct statements.
A. Compound epithelium is composed of multiple layers of cells.
B. It is mainly found in regions subjected to wear and tear.
C. Compound epithelium plays a major role in diffusion and filtration.

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D. Skin is lined by keratinised compound epithelium.

(1) A and B only
(2) A, B and D only
(3) B, C and D only
(4) A, B, C and D

31) Chondrocytes are found in :

(1) Bone
(2) Tendon
(3) Cartilage
(4) Ligament

32) Consider the following statements regarding the female reproductive system of cockroach :
A. The median oviduct opens into the genital chamber.
B. The genital chamber receives the openings of the oviduct and spermathecae.
C. The ovaries are functionally connected to the kidneys.
D. Each ovary is composed of eight ovarioles containing developing ova.
Choose the correct statements -

(1) A, B and D only
(2) A and C only
(3) B and C only
(4) All of the above

33) **Assertion (A) :** The abdomen of a cockroach consists of 10 segments.
Reason (R) : In both males and females, the 10th abdominal segment bears a pair of anal cerci.

(1) Both A and R are true, and R is the correct explanation of A.
(2) Both A and R are true, but R is not the correct explanation of A.
(3) A is true, but R is false.
(4) A is false, but R is true.

34) **Statement A :** The heart of cockroach is a long muscular tube with 13 funnel-shaped chambers.
Statement B : The haemolymph of cockroach is responsible for transporting oxygen to body tissues.

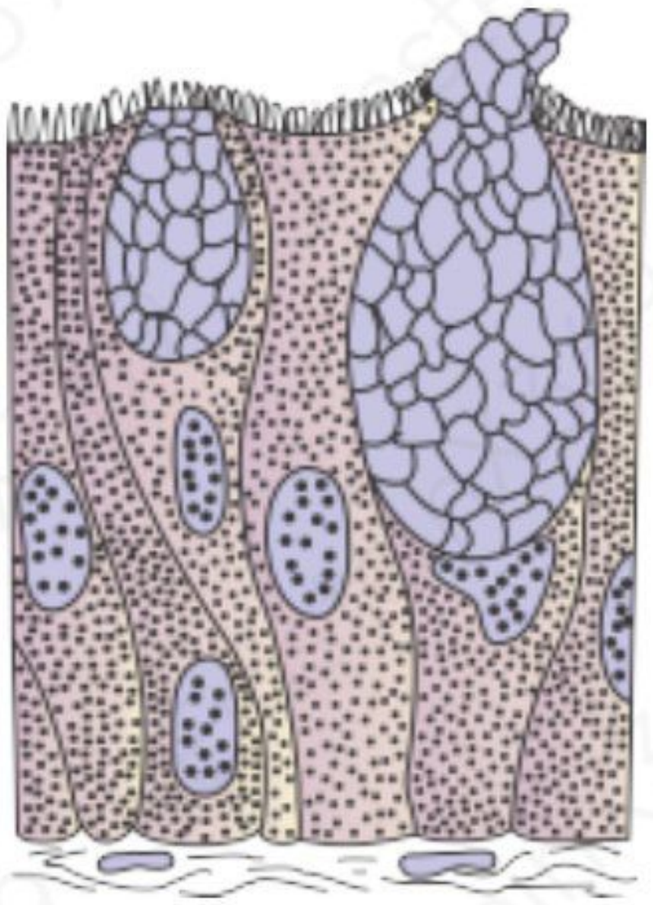
(1) Both A and B are correct.
(2) Both A and B are incorrect.
(3) A is correct but B is incorrect.
(4) A is incorrect but B is correct.

35) The young stage of cockroach that resembles the adult is called :

(1) Pupa
(2) Larva
(3) Maggot

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(4) Nymph

36) 

a

Identify the correct statement regarding (a):

(1) Unicellular gland that secretes mucus directly onto the epithelial surface.

(2) Multicellular gland with a branched duct system.

(3) Unicellular gland that releases hormones into blood.

(4) Multicellular gland without a duct.

37) Which of the following is correctly matched?

(1) Conus arteriosus — Receives blood directly from vena cava.

(2) Sinus venosus — Receives blood from major veins.

(3) Left atrium — Receives deoxygenated blood from the body.

(4) Right atrium — Opens directly into the conus arteriosus.

38) Consider the following statements regarding the male reproductive system of frog:
A. Testes are attached to the kidneys by a double fold of peritoneum called mesorchium.
B. Bidder's canal is present inside the kidney.
C. The urinogenital duct opens directly to the exterior without entering the cloaca.
D. The cloaca serves as a common chamber for the passage of faeces, urine and sperms.
Choose the correct statements -

(1) B and C only

(2) A and C only

(3) A, B and D only

(4) All of the above

39) **Assertion (A) :** Frogs are ureotelic animals.
Reason (R) : Adult frogs excrete uric acid as the chief nitrogenous waste.

(1) Both A and R are true, and R is the correct explanation of A.

(2) Both A and R are true, but R is not the correct explanation of A.

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(3) A is true, but R is false.

(4) A is false, but R is true.

40) Choose the incorrect statement -

(1) The skin of frog is smooth, moist and devoid of scales.

(2) Frogs possess mucus glands that keep the skin moist.

(3) The skin of frog acts as an accessory respiratory organ during hibernation and aestivation.

(4) The skin of frog is dry and heavily keratinised to prevent water loss.

41) **Assertion (A) :** A cell is regarded as the structural and functional unit of life because anything less than a complete cell cannot ensure independent living.
Reason (R) : Every living cell possesses membrane-bound organelles essential for survival.

(1) Both A and R are true and R is the correct explanation of A.

(2) Both A and R are true but R is not the correct explanation.

(3) A is true but R is false.

(4) Both A and R are false.

42) Which one of the following bacterial structures directly helps the organism survive hypotonic external environment?

(1) Capsule

(2) Cell wall

(3) Glycocalyx

(4) Plasma membrane

43)

Identify the correct statements.
A. Ribosomes occur in both prokaryotic and eukaryotic cells.
B. Centrosome is present in almost all plant cells.
C. Mitochondria contain circular DNA.
D. Chromatophores are characteristic membrane extensions of cyanobacteria.

(1) A, B and C

(2) A, C and D

(3) B and D only

(4) A and D only

44) Choose the correctly arranged flow showing bacterial cell envelope from outside to inside.

(1) Cell wall → Glycocalyx → Plasma membrane

(2) Plasma membrane → Cell wall → Capsule

(3) Glycocalyx → Cell wall → Plasma membrane

(4) Capsule → Plasma membrane → Cell wall

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45) Arrange the following scientists according to their contribution from earliest to latest.
i. Rudolf Virchow
ii. Robert Brown
iii. Antonie van Leeuwenhoek
iv. Matthias Schleiden

(1) iii → ii → iv → i

(2) ii → iii → iv → i

(3) iii → iv → ii → i

(4) iv → iii → ii → i

46) **Statement-I :** The fluidity of the plasma membrane is essential for endocytosis.
Statement-II : Lateral movement of membrane lipids and proteins is possible because of the quasi-fluid nature of the lipid bilayer.

(1) Both statement are true.

(2) Both statement are false.

(3) Statement-I is true but II is false.

(4) Statement-I is false but II is true.

47) Which of the following statements are correct regarding plasma membrane?
A. Cholesterol is present along with phospholipids.
B. Peripheral proteins are completely embedded inside the lipid bilayer.
C. Hydrophobic tails face towards the interior of the membrane.
D. Carrier proteins facilitate movement of certain polar molecules.

(1) A, B and D

(2) A, C and D

(3) B and C only

(4) A and C only

48) A cell is placed in a solution where only water molecules move into the cell without expenditure of ATP . The movement occurs because

(1) Water is actively transported through pumps

(2) Water diffuses from lower to higher concentration

(3) Water moves down its concentration gradient through osmosis

(4) Phospholipids transport water molecules

49) **Statement I :** The primary cell wall continues to grow during early stages of cell development.
Statement II : The secondary wall is deposited outside the primary wall.

(1) Both statements are correct.

(2) Statement I is correct but Statement II is incorrect.

(3) Statement I is incorrect but Statement II is correct.

(4) Both statements are incorrect.

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50) A student claims that all membranous organelles exchange materials directly with one another because they belong to the endomembrane system.
The incorrect part of this statement is -

(1) All membranous organelles

(2) Exchange materials

(3) Belong to the endomembrane system

(4) Both (1) and (3)

51) **Assertion (A) :** The inner mitochondrial membrane has a much larger surface area than the outer membrane.
Reason (R) : The inner membrane forms numerous infoldings called cristae.

(1) Both A and R are true and R correctly explains A.

(2) Both A and R are true but R is not the correct explanation.

(3) A is true but R is false.

(4) Both A and R are false.

52) Which of the following statements are correct regarding mitochondria?
A. Outer membrane forms the continuous boundary of the organelle.
B. Matrix contains circular DNA and 70S ribosomes.
C. Cristae are infoldings of its outer membrane.
D. Mitochondria divide by fission.

(1) A, B and D

(2) B, C and D

(3) A and C only

(4) All four

53) A mature carrot root appears orange because many of its plastids have differentiated into -

(1) Amyloplasts

(2) Chloroplasts

(3) Chromoplasts

(4) Aleuroplasts

54) **Statement I :** Chloroplast ribosomes are smaller than cytoplasmic ribosomes.
Statement II : Chloroplast ribosomes are 80S whereas cytoplasmic ribosomes are 70S.

(1) Both statement I and statement II are correct.

(2) Statement I correct, Statement II incorrect.

(3) Statement I incorrect, Statement II correct.

(4) Both statement I and statement II are incorrect.

55) Which of the following correctly matches the cellular component with its principal role according to NCERT?

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(1) Golgi — DNA replication

(2) Cytoskeleton — Mechanical support and maintenance of cell shape

(3) ER — ATP synthesis

(4) Microbodies — Ribosome assembly

56) **Assertion (A) :** The S in 80S ribosome does not represent the sum of the values of its two subunits.
Reason (R) : Svedberg unit is a measure of sedimentation coefficient and depends upon both size and density of the particle.

(1) Both A and R are true and R is the correct explanation.

(2) Both A and R are true but R is not the correct explanation.

(3) A is true but R is false.

(4) A is false but R is true.

57) A transverse section of a motile eukaryotic flagellum shows:
• Nine peripheral doublets
• One central pair
This arrangement is called -

(1) 9 + 0

(2) 9 + 1

(3) 9 + 2

(4) 8 + 2

58) Which feature is shared by both centrioles and basal bodies?

(1) Presence of central microtubule pair

(2) Nine peripheral triplets of microtubules

(3) Double membrane

(4) Ribosomes

59) Observe the given statements :
Statement-I: The outer nuclear membrane frequently bears ribosomes.
Statement-II: The outer nuclear membrane is continuous with rough endoplasmic reticulum.

(1) Both statement-I & II are true.

(2) Both statement-I & II are false.

(3) Statement-I is true but statement-II is false.

(4) Statement-I is false but statement-II is true.

60) Which structure serves as the passage for bidirectional movement of RNA and proteins between nucleus and cytoplasm?

(1) Perinuclear space

(2) Nuclear pore

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(3) Nucleolus

(4) Centromere

61) A mutation blocks all nuclear pores without damaging any other nuclear structure. Which process will be affected immediately?

(1) DNA replication

(2) Ribosomal RNA synthesis

(3) Export of mRNA to cytoplasm

(4) Transcription of DNA to mRNA

62) A chromosome possesses one arm much longer than the other because its centromere is located slightly away from the middle. It is -

(1) Metacentric

(2) Submetacentric

(3) Acrocentric

(4) Telocentric

63) **Assertion (A) :** Kinetochores are essential during cell division.
Reason (R) : Kinetochores are proteinaceous disc-shaped structures present on either side of the centromere.

(1) Both A and R are true and R is the correct explanation.

(2) Both A and R are true but R is not the correct explanation.

(3) A is true but R is false.

(4) Both A and R are false.

64) A eukaryotic cell during sprouting lacks glyoxysomes, recognize it :

(1) Castor seed cell

(2) Ground nut seed cell

(3) Wheat grain cell

(4) Mustard seed cell

65) The phenomeon responsible for disintegration of a worn off cell organelle is called as :

(1) Heterophagy

(2) Autolysis

(3) Autophagy

(4) Lysosome acting as suicidal bag

66) Pick out the correct statement.

(1) Chitin is a homopolymer.

(2) Collagen is the most abundant protein in the whole of the biosphere.

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(3) Proteins are linear chains of amino acids linked by glycosidic bonds.

(4) In a polysaccharide, the individual monosaccharides are linked by a phosphodiester bond.

67) **Assertion:** Glycosidic bonds are formed by dehydration.
Reason: In polysaccharides, individual monosaccharide is linked by glycosidic bond.

(1) Both Assertion and Reason are true, and the Reason is the correct explanation of the Assertion.

(2) Both Assertion and Reason are true, but the Reason is not the correct explanation of the Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

68) Match column I with column II and choose the correct option.

Column-I		**Column-II**	
A	Cotton fibre	I	Chitin
B	Exoskeleton of cockroach	II	Glycogen
C	Liver	III	Starch
D	Peeled potato	IV	Cellulose

(1) A – I; B – IV; C – III; D – II

(2) A – IV; B – I; C – II; D – III

(3) A – IV; B – I; C – III; D – II

(4) A – I; B – IV; C – II; D – III

69) Which of the following statements are correct?
(i) Chitin, a complex homopolysaccharide occurring in exoskeleton consists of NAG.
(ii) Glucosamine and N-acetyl glucosamine are modified sugar.
(iii) Cellulose shows blue colour when treated with I_2.
(iv) Starch in plants and glycogen in animals are store houses of energy.
(v) Right end of polysaccharide is called reducing end while left end is called non-reducing end.

(1) (i), (ii) and (iii)

(2) (iii), (iv) and (v)

(3) All except (iii)

(4) All except (v)

70) Which one of the following biomolecules is correctly characterized?

(1) Lecithin - A phosphorylated glyceride found in cell membrane.

(2) Palmitic acid - An unsaturated fatty acid with 18 carbon atoms.

(3) Adenylic acid - Adenosine with a glucose phosphate molecule.

(4) Alanine amino acid - Contains an amino group and an acidic group anywhere in the molecule.

71) In the composition of cellular mass, arrange the components- proteins(P), carbohydrates(C), lipids(L) and nucleic acids(N) in decreasing order of mass percentage.

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(1) C > N > P > L

(2) P > N > C > L

(3) P > C > L > N

(4) P > N > L > C

72) Which of the following statements about carbohydrates is/are correct?
(A) Carbohydrates are primarily composed of carbon, hydrogen, and nitrogen atoms.
(B) Starch is a polysaccharide that can hold iodine molecules in the helical portion.
(C) Inulin is a polymer of fructose.
(D) Monosaccharides are the simplest form of carbohydrates and are typically sweet tasting.

(1) A, B, and C only

(2) B, C, and D only

(3) A, C, and D only

(4) A, B, and D only

73) Which of the following statements about lipids is/are correct?
(A) Lipids are primarily composed of fatty acids and glycerol.
(B) Phospholipids are the major components of the cell membrane and have a hydrophilic head and hydrophobic tail.
(C) Lipids are soluble in water due to their polar nature.
(D) Triglycerides consist of three fatty acids bound to a single glycerol molecule.

(1) A, B, and D only

(2) A, B, and C only

(3) B, C, and D only

(4) A, C, and D only

74) The simplest carbohydrates are called _____, which include sugars like glucose and fructose. When two monosaccharides join, they form a _____, such as sucrose. Fill in the blanks with appropriate words.

(1) Disaccharides, Disaccharide

(2) Monosaccharides, Polysaccharide

(3) Disaccharides, Monosaccharide

(4) Monosaccharides, Disaccharide

75) Malonate acts as a competitive inhibitor for which of the following enzymes?

(1) Alcohol dehydrogenase

(2) Succinic dehydrogenase

(3) Catalase

(4) Amylase

76) Match column I with column II:

Column I	**Column II**

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A. Monosaccharide	i. Energy storage in animals
B. Starch	ii. Sugar units
C. Glycogen	iii. Energy storage in plants
D. Phospholipid	iv. Lecithin

(1) A–iii, B–ii, C–iv, D-i

(2) A–iii, B–iv, C–i, D-ii

(3) A–ii, B–iii, C–i, D-iv

(4) A–ii, B–i, C–iii, D–iv

77) **Statement 1:** Saturated fatty acids have at least one double bond between carbon atoms.
Statement 2: Unsaturated fatty acids contain one or more double bonds in their hydrocarbon chains, which makes them liquid at room temperature.

(1) Both statements are true.

(2) Statement 1 is false; Statement 2 is true.

(3) Statement 1 is true; Statement 2 is false.

(4) Both statements are false.

78) **Statement 1:** Cellulose is a polysaccharide found in the cell walls of plants, providing structural support.
Statement 2: Chitin, found in exoskeletons of arthropods, is a homopolymer.

(1) Both statements are true.

(2) Statement 1 is true; Statement 2 is false.

(3) Statement 1 is false; Statement 2 is true.

(4) Both statements are false.

79) Mark the statements as true or false:
a. Cell membrane and other membranes are broken into pieces, and form vesicles which are not water soluble.
b. Lipids are not strictly macromolecules.
c. Oils are solid at room temperature.
d. Triglycerides are made up of 3 fatty acid and 1 glycerol molecules.

(1) T T F T

(2) T T F F

(3) F T F T

(4) F T T F

80) How many of the following statements are true?
a. Primary metabolites usually have physiological roles.
b. Inulin is a carbohydrate while insulin is a hormone.
c. Acid insoluble fraction apart from lipids have molecular weights in the range of ten thousand Daltons and above.
d. Lipids are found in acid insoluble pool.

(1) Two

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(2) Three

(3) Four

(4) One

81) Read the following statements and mark the correct option:
a. Living organisms have got only organic elements and compounds in them.
b. Based on the nature of R group there are many amino acids. However, those which occur in proteins are only of twenty types.
c. Lipids can be monoglycerides, diglycerides and triglycerides.
d. Oils have lower melting point and hence remain as solid in winters.

(1) Only a is incorrect

(2) Only a and d are incorrect

(3) All are correct

(4) c and d are incorrect

82) **Assertion -** Polysaccharides are components of cell wall in animals, fungi and of the exoskeleton of arthropods.
Reason - Collagen is the most abundant protein in the whole of the biosphere.

(1) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.

(3) Assertion is true, but Reason is false.

(4) Both Assertion and Reason are false.

83) **Assertion (A):** Secondary metabolites such as alkaloids, flavonoids, and pigments have no known physiological role in the host organism.
Reason (R): Secondary metabolites are considered useless by-products of metabolism.

(1) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.

(3) Assertion is true, but Reason is false.

(4) Assertion is false, but Reason is true.

84) Which of the following statements about **secondary metabolites** is/are correct?
1. Secondary metabolites include compounds such as spices, alkaloids, flavonoids, rubber, and antibiotics.
2. All secondary metabolites are essential for the growth and development of the organism producing them.
3. Many secondary metabolites are useful to humans in the form of drugs, pigments, scents.
4. The ecological functions of some secondary metabolites are not yet fully understood.

(1) Statement 1, 2, and 3 are correct

(2) Statement 1, 3, and 4 are correct

(3) Statement 2, 3, and 4 are correct

(4) All four statements are correct

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85) Match column I and column II -

Column I (Compound Type)		Column II (Example)	
A	Monosaccharide	1	Glucose
B	Disaccharide	2	Sucrose
C	Polysaccharide	3	Cellulose
D	Lipid	4	Palmitic acid
E	Nucleic acid	5	DNA and RNA

(1) A–1, B–2, C–5, D–4, E–3

(2) A–1, B–3, C–2, D–4, E–5

(3) A–1, B–2, C–3, D–4, E–5

(4) A–5, B–4, C–3, D–2, E–1

86) Which of the following pairs correctly matches a **biomolecule** with the **fraction** it will be found in after trichloroacetic acid treatment?

Biomolecule	Fraction
(1) Enzyme	Acid-soluble pool
(2) RNA	Acid-insoluble fraction
(3) Glucose	Acid-insoluble fraction
(4) Starch	Acid-soluble pool

(1) 1

(2) 2

(3) 3

(4) 4

87) Which of the following statements are correct regarding elemental and biochemical composition of living organisms?

(1) Carbon is more abundant in living organisms than in the Earth’s crust.

(2) Silicon is a major component of biomolecules in living tissues.

(3) The acid-insoluble pool contains low molecular weight biomolecules.

(4) Trichloroacetic acid is used to extract lipids selectively.

88) Which of the following statements is **correct** about oils and fats?

(1) Oils are triglycerides with saturated fatty acids and high melting point.

(2) Fats are triglycerides with unsaturated fatty acids and lower melting point.

(3) Oils have more unsaturated fatty acids and remain liquid at room temperature.

(4) Fats contain phosphates and are major components of membranes.

89) Which of the following statements about proteins is FALSE?

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(1) Proteins can act as enzymes to catalyze biochemical reactions.
(2) Proteins are homopolymers made of repeating amino acids.
(3) Some proteins function as hormones.
(4) Certain proteins transport nutrients across cell membranes.

90) Which of the following **correctly matches** a protein structure level with its feature?

(1) Secondary – Disulfide bonds
(2) Primary – α-helix and β-pleated sheet
(3) Tertiary – Sequence of amino acids
(4) Quaternary – Assembly of subunits

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ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	1	2	4	1	4	1	2	3	3	2	1	1	3	2	4	2	4	4	1	4
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	2	3	2	1	2	3	2	3	2	2	1	1	1	4	1	3	1	2	3	2
Q.	41	42	43	44	45															
A.	3	4	2	2	2															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	3	1	3	3	4	2	1	4	1	1	2	2	3	4	4	4	4	4	4	2
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	1	1	3	4	1	1	2	1	4	2	3	1	3	2	4	2	3	3	3	1
Q.	86	87	88	89	90															
A.	2	1	3	1	3															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	4	3	1	2	3	1	2	3	2	1	4	4	2	4	4	3	4	3	2	2
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	1	3	1	2	4	3	3	1	3	2	3	1	2	3	4	1	2	3	3	4
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	2	2	3	1	1	2	3	2	4	1	1	3	2	2	1	3	2	1	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	3	2	1	3	3	1	2	2	3	1	2	2	1	4	2	3	2	1	1	3
Q.	171	172	173	174	175	176	177	178	179	180										
A.	2	4	3	2	3	2	1	3	2	4										

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SOLUTIONS

PHYSICS

1) **1. Question Explain :**
We need to find the exact value of sin 15º using trigonometric identities.

2. Concept Based (Sine of Angle Difference Formula)
Using the identity :
sin (A – B) = sin A cos B – cos A sin B

3. Formula used
For sin 15º = sin (45º – 30º) :
sin 15º = sin 45º cos 30º – cos 45º sin 30º
Using standard values :
$\sin 45^0 = \frac{1}{\sqrt{2}}$, $\cos 30º = \frac{\sqrt{3}}{2}$, $\cos 45º = \frac{1}{\sqrt{2}}$, $\sin 30º = \frac{1}{2}$

4. Calculation
$$\sin 15^0 = \left(\frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2}\right) - \left(\frac{1}{\sqrt{2}} \times \frac{1}{2}\right)$$
$$= \frac{\sqrt{3}}{2\sqrt{2}} - \frac{1}{2\sqrt{2}}$$
$$= \frac{\sqrt{3}-1}{2\sqrt{2}}$$
Answer :
The correct option is (1) $\frac{1}{2\sqrt{2}}\left(\sqrt{3}-1\right)$

2) **Question Explanation :** In this question we have to find the value of sin4°
Concept : This question is based on trigonometric (small angle approximation)
Solution : For small angel
$\sin(\theta) \cong \theta$ (If 'θ' is in radians)
$1° = \frac{\pi}{180}$ radian
$4° = \left(4 \times \frac{\pi}{180}\right)$ radian $= \frac{\pi}{45}$ radian
$\sin(4°) = \sin\left(\frac{\pi}{45}\right) \cong \frac{\pi}{45}$
Final answer : 2

3) **Question Explanation:**
We are given two expressions,$x_1 = 8 \sin\theta$ and $x_2 = 3 \cos\theta$. We need to determine the value of their product $x_1 x_2$ and match it with the given options.
Concept Based:
This question is based on trigonometric identities.
Formula Used:

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$2 \sin \theta \cos \theta = \sin 2\theta$

Calculation:

Solution :

$x_1x_2 = (8 \sin \theta) \times (3 \cos \theta)$

$= 24 \sin \theta \cos \theta$

Using the identity $\sin 2\theta = 2 \sin \theta \cos \theta$, we rewrite:

$$= 24 \times \frac{1}{2} \sin 2\theta$$

$= 12 \sin 2\theta$

Final Answer :

Thus, the correct answer is option (4): $12 \sin 2\theta$.

4) **Explaining :** Find slope of a straight line

$$3y = \sqrt{3}x + 3$$

Concept : $y = mx + c$ (Co-ordinate geometry)

Solution :

$3y = \sqrt{3}x + 3$

$y = \left(\frac{1}{\sqrt{3}}\right) x + 1$

$\text{Slope} = \tan\theta = \left(\frac{1}{\sqrt{3}}\right) = \tan 30^\circ$

Final Answer : (1)

5) **Explaining :** write equation of given graph

Concept : coordinate geometry

Solution :

$y = mx + c$

$y = (- \tan 45°)x + 2$

$y = - x + 2$

$x = - y + 2$

Final Answer : (4)

$$I = \int_{9}^{13} y.dx$$

6) **Explaining :** , for given x – y curve

Concept : Integration

Solution :

Area under the curve = 4×12 = 48

between x = 9 to 13

Final Answer : (1)

7) **Explaining :** Find differentiation

given $y = 3\sin\theta - 4\sin^3\theta$

find $\frac{dy}{d\theta}$ at $\theta = \frac{\pi}{3}$

Concept : differentiation

Solution :

$y = 3\sin\theta - 4\sin^3\theta = \sin(3\theta)$

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for $\theta = \frac{\pi}{3}$

$\frac{dy}{d\theta} = 3\cos(3\theta) = -3$

OR

$\frac{dy}{d\theta} = 3\cos\theta - 4 \times 3\sin^2\theta \frac{d}{d\theta}\sin\theta$

$\frac{dy}{d\theta} = 3\cos\theta - 12\sin^2\theta\cos\theta$

$\left(\frac{dy}{d\theta}\right)_{\theta=\frac{\pi}{3}} = -3$

Final Answer : (2)

8) **Explaining :** $y = \log_e x + \sin x + e^x$ then $\frac{dy}{dx}$

Concept : differentiation

Solution :

$y = \log_e x + \sin x + e^x$

$\frac{dy}{dx} = \frac{1}{x} + \cos x + e^x$

Final Answer : (3)

9) **Explaining :** $\int (3x^2 + 2x + 1)\,dx$

Concept : Integration

Solution : $\int (3x^2 + 2x + 1)\,dx = \frac{3x^3}{3} + \frac{2x^2}{2} + x + c$

$= x^3 + x^2 + x + c$ **Final Answer : (3)**

10) **Explaining :** $1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \ldots\ldots \infty =$

Concept : Geometric progression

$S_\infty = \frac{a}{1-r}$

Solution : $S_\infty = \frac{a}{1-r}$

$S_\infty = \frac{1}{1-\frac{1}{3}} = \frac{3}{2}$ **Final Answer : (2)**

11) **Explaining :** To find the magnitude of the scalar product.

Concept : This question is based on concept of dot product.

Solution : $\vec{F}_1 \cdot \vec{F}_2 = (5 \times 4) = 20$

Final Answer : (1)

12)

Explain Question :- In this question we are asked to find the angle between two $\vec{A}$ vectors and $\vec{B}$.

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Concept :-This question is based on Dot product of two vectors

Solution :- The dot product of two vector $\vec{A}$ and $\vec{B}$ is given by

$\vec{A}.\vec{B} = AB\cos\theta$

$\Rightarrow \cos\theta = \frac{\vec{A}.\vec{B}}{AB}$

given here

$\vec{A}.\vec{B} = \sqrt{3}$, A = 1, B = 2

$\therefore \cos\theta = \frac{\sqrt{3}}{1.2} = \frac{\sqrt{3}}{2} \Rightarrow \therefore \theta = 30°$

Final answer :- 1

13) **Explaining :** $\hat{a}$ and $\hat{b}$ are non-collinear unit vectors and $\left|\hat{a}+\hat{b}\right| = \sqrt{3}$

Concept : dot product

Solution : $\left|\hat{a}+\hat{b}\right| = \sqrt{3}$

$\sqrt{1^2+1^2+2.1.1\ \cos\theta} = \sqrt{3}$

$\Rightarrow \quad 2\cos\theta = 1 \Rightarrow \cos\theta = \frac{1}{2} \Rightarrow \theta = 60°$

$(\hat{a}-5\hat{b})\ .\ (3\hat{a}+\hat{b})$

$= 3(\hat{a}\ .\ \hat{a}) + (\hat{a}\ .\ \hat{b}) - 15\,(\hat{b}\ .\ \hat{a}) - 5(\hat{b}\cdot\hat{b})$

$= 3 + |\hat{a}||\hat{b}|\cos 60° - 15|\hat{b}||\hat{a}|\cos 60° - 5$

$= 3 + \frac{1}{2} - \frac{15}{2} - 5$

$= -2 - 7 = (-9)$ **Final Answer : (3)**

14) **Explaining :** If a vector $2\hat{i}+3\hat{j}+8\hat{k}$ is perpendicular to the vector $-4\hat{i}+4\hat{j}+\alpha\hat{k}$.

Concept : Application of dot product

Solution : $\left(2\hat{i}+3\hat{j}+8\hat{k}\right).\left(-4\hat{i}+4\hat{j}+\alpha\hat{k}\right) = 0$

$-8 + 12 + 8\alpha = 0 \Rightarrow \alpha = -\frac{1}{2}$ **Final Answer : (2)**

15) **Explaining :** The vector $\vec{B} = 5\hat{i}+2\hat{j}-S\hat{k}$ is perpendicular to the vector $\vec{A} = 3\hat{i}+\hat{j}+2\hat{k}$

Concept : Application of dot product

Solution :

If $\vec{A}\perp\vec{B}$ then $\vec{A}.\vec{B} = 0$

$\left(3\hat{i}+\hat{j}+2\hat{k}\right).(5\hat{i}+2\hat{j}-S\hat{k}) = 0$

$\Rightarrow \quad 15 + 2 - 2S = 0$

$\Rightarrow \quad 2S = 17 \quad \Rightarrow \quad S = 8.5$

Final Answer : (4)

16) **Question Explanation:** In this question we have to find the angle $\vec{A}$ and $\vec{B}$ if $\left|\vec{A}\times\vec{B}\right| = \frac{AB}{2}$

Concept: Application of cross product

Solution: By the formula of cross product of two vectors

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$$\left|\vec{A}\times\vec{B}\right| = AB\sin\theta$$

$$(\vec{A}\times\vec{B}) = \frac{AB}{2}$$ (Given in the questions)

$$AB\sin\theta = \frac{AB}{2}$$

$$\sin\theta = \frac{1}{2}$$

$$\therefore\theta = 30^\circ \quad \left[\because \sin 30^\circ = \frac{1}{2}\right]$$

Final Answer: Option (2)

17) **Explaining :** $(\vec{A}+\vec{B})\times(\vec{A}-\vec{B})$

Concept : cross product

Solution :

$$\left(\vec{A}+\vec{B}\right)\times\left(\vec{A}-\vec{B}\right) = \vec{A}\times\vec{A}-\vec{A}\times\vec{B}+\vec{B}\times\vec{A}-\vec{B}\times\vec{B}$$

$$= \vec{B}\times\vec{A}+\vec{B}\times\vec{A} = 2\left(\vec{B}\times\vec{A}\right)$$ **Final Answer : (4)**

18) **Explaining :** $\vec{A}\times\vec{B} = \vec{C}$

Concept : Cross product

Solution :

$\vec{A}\times\vec{B} = \vec{C}$

$\vec{C}$ is $\perp^{ar}$ to $\vec{A}$ as well as $\vec{B}$ and also $\left(\vec{A}+\vec{B}\right)$

Final Answer : (4)

19) **Explaining :** A = 5 units, B = 6 units and $|\vec{A}\times\vec{B}| = 15$ units

Concept : cross product

Solution :

$$\sin\theta = \frac{\left|\vec{A}\times\vec{B}\right|}{AB} = \frac{15}{5\times 6} = \frac{1}{2} \Rightarrow \theta = 30^\circ$$

Final Answer : (1)

20) **Explaining :** $\vec{A} = 2\hat{i}+3\hat{j}$ and $\vec{B} = \hat{i}+4\hat{j}$

Concept : Area of parallelogram

Solution :

$$\left|\vec{A}\times\vec{B}\right| = \left|\left(2\hat{i}+3\hat{j}\right)\times\left(\hat{i}+4\hat{j}\right)\right| = \left|5\hat{k}\right| = 5 \text{ units}$$

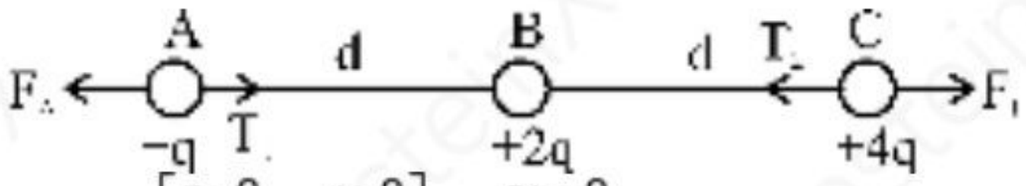
Final Answer : (4)

21) **Explaining :** Three charges +q, +2q and +4q are connected by strings as shown in the figure. What is the ratio of tensions in the strings AB and BC.

Concept : This question is based on in equilibrium $\sum\vec{F_x} = 0$

Solution :

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$$T_1 = k\left[\frac{2q^2}{d^2} + \frac{4q^2}{4d^2}\right] = \frac{3kq^2}{d^2}$$

$$T_2 = k\left[\frac{8q^2}{d^2} + \frac{4q^2}{4d^2}\right] = \frac{9kq^2}{d^2}$$

$$\frac{T_1}{T_2} = \frac{1}{3}$$

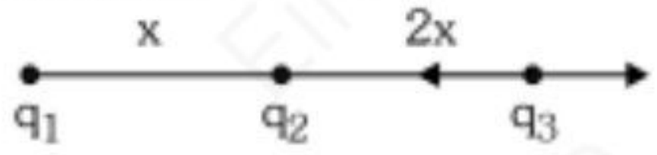
Final Answer : (2)

22) **Explaining :** Three charged particles are placed on a straight line as shown in the figure. q_1 and q_2 are fixed but q_3 can move. Under the action of the forces from q_1 and q_2 , q_3 is in equilibrium. What is the relation between q_1 and q_2 ?

Concept : This question is based on for equilibrium of q_3

$Fq_3=0$

Solution :



-For q_3 in equilibrium net force = 0

$$\frac{Kq_1q_3}{9x^2} = -\frac{Kq_2q_3}{4x^2}$$

$$q_1 = -\frac{9}{4}q_2$$

Final Answer : (3)

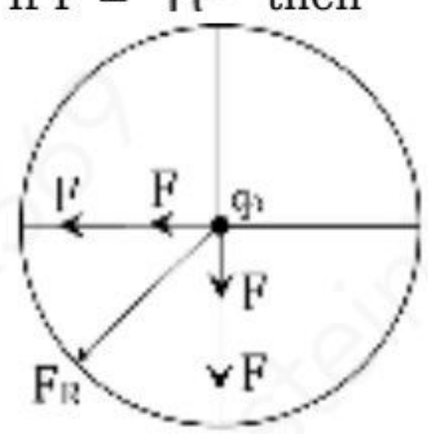
23)

Explanation Question: We have to find the resultant force on a charge q_0 at center what time shows by the clock

Concept: This question is based on electrostatic force

Solution:

If $F = \frac{kqq_0}{R^2}$ then



Time on clock will be 7 : 30

Final Answer : 7 : 30 will be answer

24) **Explaining :** discuss the type of equilibrium

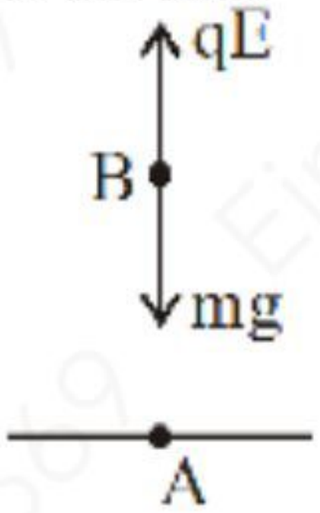
Concept : This question is based on equilibrium

Solution :

If B is pushed down, upwards electrostatic repulsion on it increases and hence it return back

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& vice versa



Final Answer: Option(1)

25)

Question Explanation : Two bodies are rubbed with each other, one gets positive charge and the other gets negative charge. we need to identify the body having higher work function.

Concept : Methods of Charging of two bodies **(**Work function & electron transfer)

Solution

Work function is the **minimum amount of energy required to remove an electron from the surface of a material**.
On rubbing, **electrons transfer** from one body to the other.
The body that **loses electrons** becomes **positively charged** → it has **lower work function** (electrons escape easily).
The body that **gains electrons** becomes **negatively charged** → it has **higher work function** (holds electrons tightly).

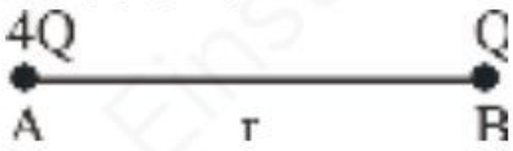
Given:
A becomes **positive** ⇒ A lost electrons ⇒ **lower work function**
B becomes **negative** ⇒ B gained electrons ⇒ **higher work function**

Final Answer : option (2)

26) **Explaining :** Force between A and B is F. If 75% charge of A is transferred to B then force between A and B is :

Concept : This question is based on redistribution of charge

Solution :



$$\Rightarrow F = \frac{k(4Q)(Q)}{r^2} = \frac{4kQ^2}{r^2}$$



$$\Rightarrow F^1 = \frac{k(4Q)(Q)}{r^2} = F$$

Final Answer : (3)

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27) **Explaining the Question :-** Asking about the ratio of force $|\vec{F}_1|$ to $|\vec{F}_2|$ Before & after the connecting the spheres changed conducting

Concept:- This question is based on coulombs low

Solution :-

 coulombs low is related as $F \propto q, q_2$

$$\frac{|F_1|}{|F_2|} = \frac{(10\mu C)(20\mu C)}{(5\mu C)(5\mu C)} \Rightarrow \frac{8}{1}$$

Final Answer :- (2)

28) **Question Explanation:**

The question asks to find the magnitude of the electrostatic force on the charge Q placed at a vertex of an equilateral triangle, where the other two vertices have charges -Q and -Q, and the side length of the triangle is a.

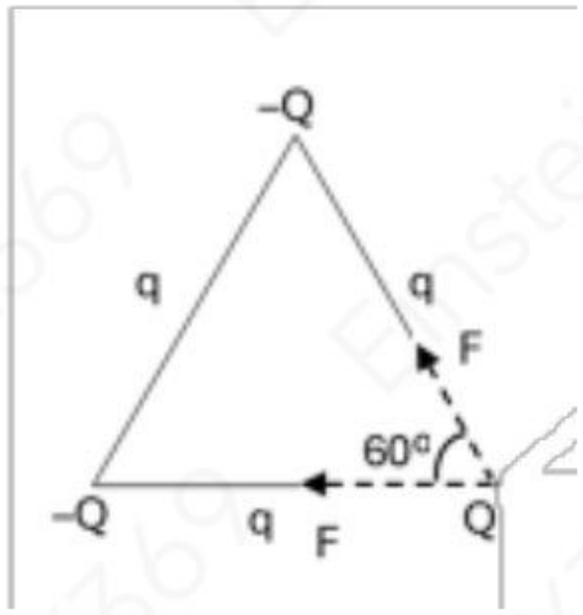
Concept :

Coulomb's Law and superposition of forces.

Solution : Coulomb's Law is used to calculate the electrostatic force between two charges:

$$F = k\frac{|q_1 q_2|}{r^2}$$

$$F_1 = k\frac{|Q \times (-Q)|}{a^2} = k\frac{Q^2}{a^2}.$$



$$F_2 = k\frac{|Q \times (-Q)|}{a^2} = k\frac{Q^2}{a^2} \text{ as } |F_1| = |F_2| = F \text{ (let)}$$

Angle between two equal vectors is 60° then their resultant is

$$F_{resultant} = F\sqrt{3}$$

$$= \frac{\sqrt{3}KQ^2}{a^2}$$

Final Answer : Option (3)

29)

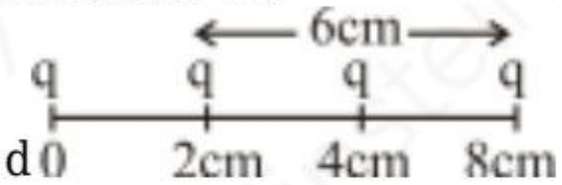
Question Explanation : Four identical charges of 2 μC are placed on the x-axis at x = 0 cm, 2 cm, 4 cm, and 8 cm. We have to find the net force exerted on the charge placed at x = 2 cm.

Concept : This question is based on Coulomb's Law and superposition of Forces

Formula : $F = \frac{kq_1 q_2}{r^2}$

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Solution : d



Distance from charge placed at 0 cm = 2 cm → force towards +x

Distance from charge placed at 4 cm = 2 cm → force towards –x

These two forces are equal in magnitude and opposite in direction, so they cancel each other.

Distance from charge placed at 8 cm = 6 cm = 0.06

Net force = force due to charge at 8 cm only

$$F = \frac{\left(9 \times 10^{9}\right)\left(2 \times 10^{-6}\right)^{2}}{(0.06)^{2}}$$

$$F = \frac{\left(9 \times 10^{9}\right)\left(4 \times 10^{-12}\right)}{0.0036}$$

$$F = \frac{0.036}{0.0036}$$

N = 10 N

Final Answer : Option (2)

30) **Explaining :** Two uniformly charged rings of radius R and 2R are placed co-axially as shown in figure. If net electric field at point P, which is x (x < < R) distance away from common centre 'O' is zero then which relation is correct for charge in rings

Concept : This question is based on electric field due to ring

Solution :

Electric field due to Q_1 = – (Electric field due to Q_2)

$$\Rightarrow \frac{KQ_1x}{\left(R^2+x^2\right)^{3/2}} = -\frac{KQ_2x}{\left((2R)^2+x^2\right)^{3/2}}$$

 x < < R

than $x^2 \approx O$

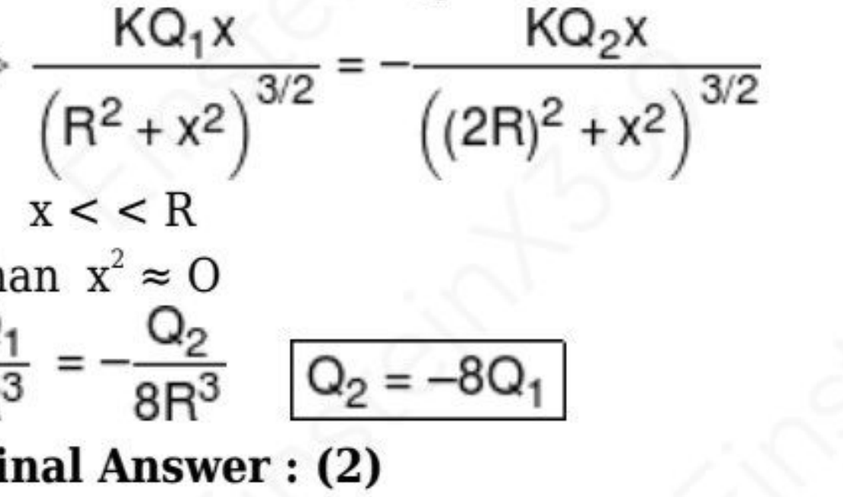
$$\frac{Q_1}{R^3} = -\frac{Q_2}{8R^3} \qquad \boxed{Q_2 = -8Q_1}$$

Final Answer : (2)

31) **Explaining :** The charge per unit length of the four quadrant of the ring is 2λ, – 2λ, λ and – λ respectively. The magnitude of electric field at the centre is :

Concept : This question is based on electric field due to arc

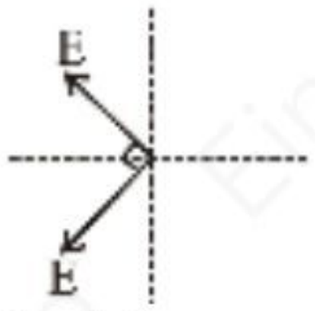
Solution :



$$E = \frac{\lambda}{2\pi\in_0 r}\sin\frac{\theta}{2} = K\frac{2\lambda}{r}\sin\frac{\pi}{4}$$

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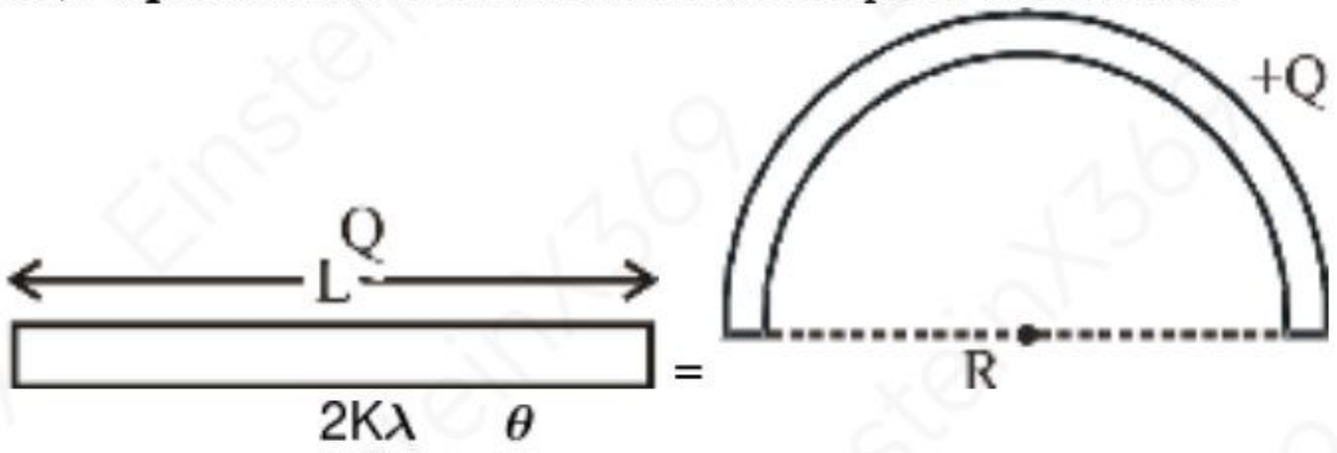
For $\theta = \pi/2$, $E = \frac{K\lambda\sqrt{2}}{r}$



$E_{net} = E\sqrt{2} = \frac{K\lambda 2}{r}$

Final Answer : (1)

32) **Explanation:** Rod is bent in the shape of a semicircle



$= \frac{2K\lambda}{R}\sin\frac{\theta}{2}$

Concept: $E = \frac{2K\lambda}{R}\sin\frac{\theta}{2}$

Solution:

$E = \frac{2K\lambda}{R} = \frac{\frac{2\times 1}{4\pi\in_0}\frac{Q}{L}}{L\times\frac{L}{\pi}}$ [πR = L]

$$\boxed{E = \frac{Q}{2\in_0 L^2}}$$

Final Answer: (1)

33) **Explanation:** V vs r is non conducting uniformly charged sphere.

Concept: $E = \frac{\rho}{3\in_0}r$

Solution:

$\because E = \frac{\rho}{3\in_0}r \therefore \frac{\rho}{3\in_0} = \frac{4}{3} \Rightarrow \rho = 4\in_0$

$Q = \rho$ [volume] $= 4\in^0\left[\frac{4}{3}\pi 27\right] = 144\ \pi\in^0$

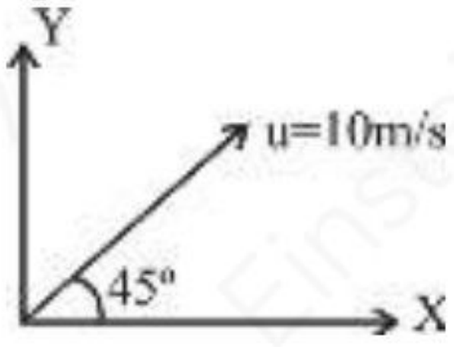
Final Answer: (1)

34) **Explanation:** Motion under uniform electric and gravitational field

Concept: $a_x = \frac{qE}{m}$ and $a_y = -g$

Solution:

$a_x = \frac{qE}{m} = \frac{10^{-6}\times 2\times 10^7}{2} = 10\ m/s^2$



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Time of flight $T = \frac{2u\sin\theta}{g} = \frac{2 \times 10 \times \frac{1}{\sqrt{2}}}{10} = \sqrt{2}s$

Hence, Range $R = u_xT + \frac{1}{2}a_xT^2$

$R = 10\cos 45° \times T + \frac{1}{2}a_xT^2$

$\frac{10}{\sqrt{2}} \times \sqrt{2} + \frac{1}{2} \times 10 \times 2 = 20m$

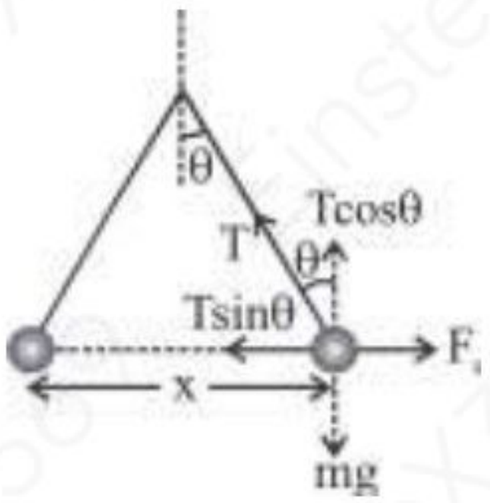
(use g = 10 m/s^2)

Final Answer: (4)

35) **Explanation:** Equilibrium of suspended charge system

Concept: In equilibrium $\tan\theta = \frac{F_e}{mg}$

Solution:



In equilibrium $F_e = T\sin\theta$ ...(i)

$mg = T\cos\theta$ ...(ii)

$\tan\theta = \frac{F_e}{mg} = \frac{q^2}{4\pi\varepsilon_0 x^2 \times mg}$

Also $\tan\theta \approx \sin\theta = \frac{x/2}{L}$

Hence $\frac{x}{2L} = \frac{q^2}{4\pi\varepsilon_0 x^2 \times mg}$

$\Rightarrow x^3 = \frac{2q^2L}{4\pi\varepsilon_0 mg} \Rightarrow x = \left(\frac{q^2L}{2\pi\varepsilon_0 mg}\right)^{1/3}$

Final Answer: (1)

36) **Question explain:** This explores the linear relationship between the amount of charge packed into a volume and the resulting force field generated at an internal point.

Concept : Field-Density Relationship In a static distribution, the electric field is directly proportional to the charge density ρ as per the differential form of Gauss's Law.

Formula used: $E \propto \rho$

Calculation: Since $E = \rho r / (3\varepsilon_0)$, if $\rho' = 2\rho$, then $E' = 2\rho r / (3\varepsilon_0) = 2E$.

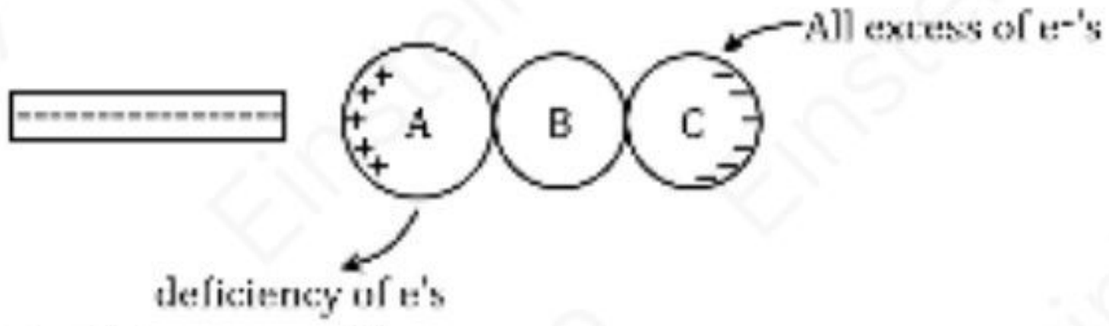
Final Answer: Option (3)

37) **Explanation:** Given a charge rod near neutral body

Concept: Electrostatics induction

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Solution:



Final Answer: (1)

38) **Explanation:** Charge distributed on circular arc

Concept: $V = \frac{kQ}{R}$

Solution:

$$V = \frac{kQ}{R} + \frac{k(3Q)}{R} + \frac{k(-2Q)}{R}$$

$$= \frac{2kQ}{R} = \frac{Q}{2\pi\varepsilon_0 R}$$

Final Answer: (2)

39) **Explanation:** Given dipole in non-uniform electric field.

Concept: $\vec{F} = q\vec{E}$

Solution:

E_{ext} at $-q > E_{ext}$ at $(+q)$

Final Answer: (3)

40) **Explanation:** Asked plux passing through the cylinder

Concept: $\phi = \frac{q_{enclosed}}{\varepsilon_0}$

Solution:

Charge enclosed = $q = \int_0^L \lambda dy$

$$= \int_0^L (\lambda_0 y)\, dy = \lambda_0 \left(\frac{L^2}{2}\right)$$

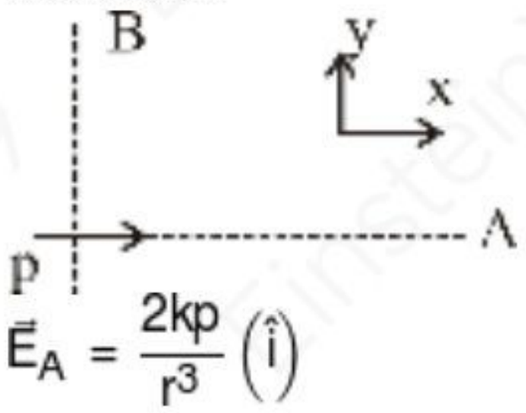
So $\phi_e = \frac{\lambda_0 L^2}{2\epsilon_0}$

Final Answer: (2)

41) **Explanation:** Find field at point A and B

Concept: $\vec{E}_A = \frac{2kp}{r^3}(\hat{i})$, $\vec{E}_B = \frac{kp}{r^3}(-\hat{i})$

Solution:



$$\vec{E}_A = \frac{2kp}{r^3}(\hat{i})$$

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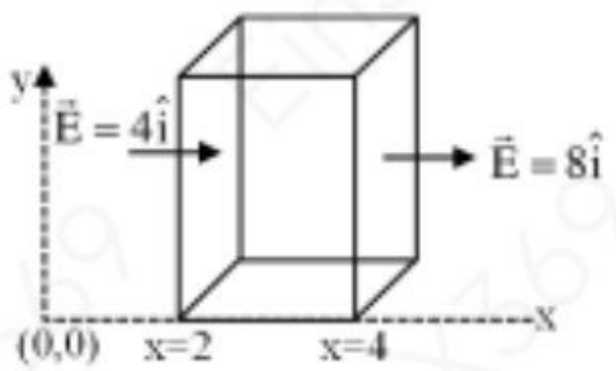
$$\vec{E}_B = \frac{kp}{r^3}\left(-\hat{i}\right) \Rightarrow \vec{E}_A = -2\vec{E}_B$$

Final Answer: (3**)**

42) **Explanation:** Given: two parallel metallic plates, maintained at different potential.
Asked: The motion of an electron if it is released midway between the plates.
Concept: The direction Electric field is from higher potential of –200 V to lower potential of –400 V (right ward). The electron accelerates opposite to the electric field.
Solution: Electric field is direction right ward (higher potential of –200 V to lower potential of –400 V). When electron left free in an electric it accelerates opposite to the electric field. Hence in the given case electron accelerates left ward.
Final Answer: (4)

43) **Explanation:** Given: Force between two point charges q_1 and q_2 placed in vacuum at 'r' cm apart is F.
Asked: The force between them when placed in a medium having dielectric K = 5 at 'r/5' cm apart.
Concept: $F = \frac{1}{4\pi \in_0}\frac{q_1 q_2}{r_2}$
Solution:
In air $F = \frac{1}{4\pi \in_0}\frac{q_1 q_2}{r_2}$
In medium $F' = \frac{1}{4\pi (K \in_0)}\frac{q_1 q_2}{(r')^2} = \frac{25}{4\pi (5 \in_0)}\frac{q_1 q_2}{(r)^2} = 5F$
Final Answer : (2)

44) **Explanation:** Given: An electric field $\vec{E} = (2x\hat{i})NC^{-1}$ exists in space.
Asked: The electric flux through the cube.
Concept: $\phi = \vec{E}.\vec{A}$
Solution:



$\vec{E} = 2x\hat{i}$
$\phi = E.A$
$\phi_{in} = -4 \times 4 = -16\,Nm^2/c$
$\phi_{out} = 8 \times 4 = 32Nm^2/c$
$d_{net} = \phi_{in} + \phi_{out} = -16 + 32 = 16\ Nm^2/c$
Final Answer : (2)

45) **Explaining :** To find dimension of given quantity.
Concept : This question is based on $V = \frac{W}{q}$

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Solution : $[V] = \frac{M^1L^2T^{-2}}{AT}$
Final Answer : (2)

CHEMISTRY

46) **Explaining :** Which statement about molecularity of a reaction is wrong ?
Concept : This question is based on molecularity.
Solution : Molecularity can never be fractional.
Final Answer : (3)

47) **Explaining :** Higher order (> 3) reactions are rare due to:
Concept : This question is based on order of reaction.
Solution : low probability of simultaneous collision of all the reacting species
Final Answer : (1)

48) **Explaining :** The rate constant of nth order has unit:
Concept : This question is based on unit of rate constant.

Solution : For nth order; unit of rate constant may be derived by $k = \frac{rate}{[reactant]^n}$.
Final Answer : (3)

49) **Explaining :** Plots showing the variation of the rate constant (k) with temperature (T) are given below. The plot that follows Arrhenius equation is:
Concept : This question is based on Arrhenius equation.
Solution : $k = Ae^{-E_a/RT}$ i.e., k increase exponentially, with rise in temperature.
Final Answer : (3)

50) **Explaining :** For producing the effective collisions the colliding molecules must have:
Concept : This question is based on collision theory.
Solution : This is activation state and orientation concept for mechanism of reactions.
Final Answer : (4)

51) **Explaining :** Rate law expression.
Concept : This question is based on rate law for complex reactions.
Solution :
Slowest step of mechanism decides the rate expression,
Thus, rate = k $[NO_2]$ $[F_2]$
Final Answer : (2)

52) **Explaining :** For a reaction for which the activation energies of forward and reverse reactions are equal ?
Concept : This question is based on enthalpy of reaction.
Solution :

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If energy of activation for forward and backward reactions are same, reaction is neither exothermic nor endothermic.
Final Answer : (1)

53) **Explaining :** The differential rate expression for the reaction $H_2 + I_2 \rightarrow 2HI$ is:
Concept : This question is based on rate of reaction in terms of stoichiometry.
Solution : $\frac{-d[H_2]}{dt} = \frac{-d[I_2]}{dt} = \frac{1}{2}\frac{d[HI]}{dt}$
Final Answer : (4)

54) **Explaining :** For the reaction $N_2 + 3H_2 \rightarrow 2NH_3$, $\frac{d[NH_3]}{dt} = 2 \times 10^{-4} Ms^{-1}$. Therefore, the rate $-\frac{d[N_2]}{dt}$ is given as:
Concept : This question is based on rate of reaction.
Solution : $\frac{-d[N_2]}{dt} = \frac{1}{2}\frac{d[NH_3]}{dt}$
Final Answer : (1)

55) **Explaining :** The relation in between k_1, k_2 and k_3 is :
Concept : This question is based on rate in terms of stoichiometry of reaction.
Solution :
$\frac{-d[N_2O_5]}{dt} = \frac{1}{2}\frac{d[NO_2]}{dt} = \frac{2d[O_2]}{dt}$,
$\therefore k_1[N_2O_5] = \frac{k_2}{2}[N_2O_5] = 2k_3[N_2O_5]$
Final Answer : (1)

56) **Explaining :** Compounds 'A' and 'B' react according to the following chemical equation.
Concept : This question is based on rate law.
Solution : Solve using rate = $k [A]^m[B]^n$
Final Answer : (2)

57) **Explaining :** The rate of reaction between two reactants A and B decreases by a factor 4 if the concentration of reactant B is doubled. The order of this reaction with respect to B is :
Concept : This question is based on order of reaction.
Solution : $r = k' [A]^m[B]^n$
Also, $\frac{r}{4} = k'[A]^m[2B]^n$
$4 = \left(\frac{1}{2}\right)^n$ or $2^2 = 2^{-n}$
$\therefore$ $n = -2$
Final Answer : (2)

58) **Explaining :** The rate constant k_1 and k_2 for two different reactions are $10^{16}\ e^{-2000/T}$ and $10^{15}\ e^{-1000/T}$, respectively. The temperature at which $k_1 = k_2$ is :
Concept : This question is based on Arrhenius equation.

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Solution :

$K_1 = 10^{16}\, e^{-2000/T}$; $K_2 = 10^{15}\, e^{-1000/T}$

if $K_1 = K_2$ then $10^{16}\, e^{-2000/T} = 10^{15}\, e^{-1000/T}$

or $\log 10 - \frac{2000}{T} = \frac{1000}{T}$ or $T = \frac{1000}{2.303} K$

Final Answer : (3)

59) **Explaining :** Half-life period of a first order reaction is 1386 seconds. The specific reaction rate of the reaction is :

Concept : This question is based on first order reactions.

Solution : For first order, $k = \frac{0.693}{t_{1/2}} = \frac{0.693}{1386} = 5 \times 10^{-4} s^{-1}$

Final Answer : (4)

60) **Explaining :** In a zero-order reaction for every 10°C rise of temperature, the rate is doubled. If the temperature is increased from 10°C to 100°C, the rate of the reaction will become:

Concept : This question is based on effect of temperature on rate of reaction.

Solution :

$\frac{r(t+10)}{r_t} = 2$ for each 10°C rise in temperature

$\frac{r_{100}}{r_{10}} = (2)^9 = 512$

Final Answer : (4)

61) **Explaining :** Activation energy (E_a) and rate constants (k_1 and k_2) of a chemical reaction at two different temperatures (T_1 and T_2) are related by :

Concept : This question is based on Arrhenius equation.

Solution :

On integrating within limits k_1 and k_2 and T_1 to T_2

$$\int_{k_1}^{k_2} \ln K = -\frac{E_a}{R} \int_{T_1}^{T_2} \frac{1}{T}$$

$$\ln \frac{k_2}{k_1} = -\frac{E_a}{R}\left[\frac{1}{T_2} - \frac{1}{T_1}\right] = \frac{E_a}{R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$$

or $\ln \frac{k_2}{k_1} = \frac{E_a}{R}\left[\frac{T_2 - T_1}{T_1 T_2}\right]$ or $\ln \frac{k_2}{k_1} = \frac{-E_a}{R}\left[\frac{1}{T_2} - \frac{1}{T_1}\right]$

Final Answer : (4)

62) **Explaining :** When initial concentration of a reactant is doubled in a reaction, its rate is not affected. The order of the reaction is :

Concept : This question is based on zero order reaction.

Solution : $r \propto [A]^\circ$ for zero order reaction.

Final Answer : (4)

63) **Explaining :** Find relation between $(ROD)_{H_2}$ and $(ROA)_{NH_3}$.

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Concept : This question is based on rate of reaction.
Solution : Rate of reaction = $-\frac{1}{3}\frac{d[H_2]}{dt} = -\frac{d[N_2]}{dt} = \frac{1}{2}\frac{d[NH_3]}{dt}$
Final Answer : (4)

64) **Explaining :** Determine rate law.
Concept : This question is based on rate law.
Solution : Use : $r = k\,[A]^m[B]^n$
Final Answer : (4)

65) **Explaining :** Evaluate statement and explanation.
Concept : This question is based on collision theory.
Solution : Only those molecules react which acquire threshold energy level & have proper orientation
Final Answer : (2)

66) **Explaining :** In a first order reaction A → B, if k is the rate constant and initial concentration of the reactant is 0.5 M, then half-life is :
Concept : This question is based on first order reaction.
Solution :
For I order reaction, $k = \frac{2.303\ \log_{10}2}{t_{1/2}}$
$t_{1/2} = \frac{0.693}{k} = \frac{\ln 2}{k}$
Final Answer : (1)

67) **Explaining :** Activation energy of a chemical reaction can be determined by :
Concept : This question is based on Arrhenius equation.
Solution : By, 2.303 $\log\frac{k_2}{k_1} = \frac{E_a}{R}\left[\frac{T_2 - T_1}{T_1T_2}\right]$
Final Answer : (1)

68) **Explaining :** In the respect of the equation $k = Ae^{-E_a/RT}$ in chemical kinetics, which one of the following statements is correct
Concept : This question is based on Arrhenius equation.
Solution :
$\ln k = Ae^{-E_a/RT}$, Ea is energy of activation, k is rate constant, A is Arrhenius factor, R is universal gas constant.
Final Answer : (3)

69) **Explaining :** Normality of 10% (w/v) H_2SO_4 solution is nearly :
Concept : This question is based on normality.
Solution :
$N = \frac{w \times 1000}{E_A \times V_{mL}} = \frac{10 \times 1000}{49 \times 100}$
$N = 2$

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Final Answer : (4)

70) **Explaining :** 1000 g $CaCO_3$ solution contains 10 g carbonate. The concentration of solution is :
Concept : This question is based on concentration in terms of ppm.
Solution : $\text{ppm} = \frac{10}{1000} \times 10^6 = 10^4$
Final Answer : (1)

71) **Explaining :** On mixing 10 mL of carbon tetrachloride with 10 mL of benzene both being non-polar the total volume of the solution is:
Concept : This question is based on ideal solutions.
Solution : A mixture of CCl_4 and C_6H_6 shows a positive of deviation from Raoult's law.
Final Answer : (1)

72) **Explaining :** The value of K_b for water is 1.86, calculated from Glucose solution. The value of K_b for water calculated for NaCl solution will be:
Concept : This question is based on molal elevation constant.
Solution : K_b depends on nature of solvent.
Final Answer : (2)

73) **Explaining :** What is the normal freezing point of the solution represented by the phase diagram ?
Concept : This question is based on depression in freezing point.
Solution :
Freezing point is the temperature at which the liquid and the solid state of a substance have the same vapour pressure.
 The normal freezing point of the solution is T_1, because in the given graph it is the temperature at which the solid and solution curves intersect at 1 atm pressure.
Final Answer : (1)

74) **Explaining :** Calculate P_A^0 and P_B^0.
Concept : This question is based on Raoult's law.
Solution :
$P = P_A^0 X_A + P_B^0 X_B$
$P = P_A^0 X_A + P_B^0 (1 - X_A)$
$= P_A^0 X_A + P_B^0 - P_B^0 X_A$
$P = P_B^0 - X_A (P_B^0 - P_A^0)$
$P = 254 - 119\ X_A$ on comparing
$P_B^0 = 254$ and $P_B^0 - P_A^0 = 119$
then $P_A^0 = 135$
Final Answer : (4)

75) **Explaining :** An ideal solution contains two volatile liquids A (P^0 = 100 torr) and B (P^0 =

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200 torr). If mixture contain 1 mole of A and 4 mole of B then total vapour pressure of the solution is:

Concept : This question is based on Raoult's law.

Solution :

$$P = P_A^0 X_A + P_B^0 X_B$$

$$= 100 \times \frac{1}{5} + 200 \times \frac{4}{5}$$

$$= 180 \text{ torr}$$

Final Answer : (2)

76) **Explaining :** Solution of two liquids A and B showing negative deviation from Raoult's law?

Concept : This question is based on Raoult's law.

Solution :

In non ideal solutions showing negative deviation from Raoult's ; the A-B interactions are stronger than the A-A and B-B molecular interactions.

$$P_A < P_A^0 X_A; P_B < P_B^0 X_B$$

$$P_{total} < P_A^0 X_A + P_B^0 X_B$$

$$\Delta H_{mix} < 0$$

$$\Delta V_{mix} < 0$$

$$\Delta S_{mix} > 0$$

Final Answer : (3)

77) **Explaining :** Two liquids having vapour pressures P_1^0 and P_2^0 in pure state in the ratio of 2 : 1 are mixed in the molar ratio of 1 : 2. The ratio of their moles in the vapour state would be:

Concept : This question is based on Raoult's law.

Solution :

$$Y_A = \frac{P_A^0 X_A}{P} \quad \ldots (i)$$

$$Y_B = \frac{P_B^0 \times X_B}{P} \quad \ldots (ii)$$

$$\text{Ratio } \frac{(i)}{(ii)} \quad \frac{Y_A}{Y_B} = \frac{P_A^0}{P_B^0} \times \frac{X_A}{X_B}$$

$$= \frac{2}{1} \times \frac{1}{2} = 1 : 1$$

Final Answer : (1)

78) **Explaining :** Match the Column-I with Column-II :

Concept : This question is based on concentration

Solution :

(A) Molality → (r) Elevation in boiling point is proportional to it

(B) Molarity → (q) Varies with temperature

(C) Mole fraction of solute →(p) Equal to the relative lowering of vapour pressure

(D) Normality → (q) Varies with temperature, (s) Number of equivalents of solute dissolved per litre of solution

Final Answer : (3)

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79) **Explaining :** Vapour pressure of a solution is:
Concept : This question is based on vapour pressure.
Solution : Vapour pressure of solvent in the solution is directly proportional to mole fraction of solvent in the solution.
Final Answer : (2)

80) **Explaining :** Calculate the mass of a non-volatile solute (molar mass 40 g mol^{-1}) which should be dissolved in 114 g octane to reduce its vapour pressure to 80%:
Concept : This question is based on relative lowering of vapour pressure.
Solution :

P_s = 80% of P^0 = 0.80 P^0 . Solute = $\frac{w}{40}$ mol

Solvent (octane) = $\frac{114\ g}{114\ g\ mol^{-1}} = 1$

Molar mass of C_8H_{18} = 114 g mol^{-1}

Now, $\frac{P^0 - P_s}{P^0} = X_2$

$\therefore \frac{P^0 - 0.80P^0}{P^0} = \frac{w/40}{w/40 + 1}$

$0.2\left(\frac{w}{40} + 1\right) = \frac{w}{40}$ or $\frac{0.80w}{40} = 0.2$ or w = 10 g

Final Answer : (4)

81) **Explaining :** Decrease in the vapour pressure of the solvent depends on:
Concept : This question is based on lowering of vapour pressure.
Solution : Decrease in the vapour pressure of the solvent depends on the quantity of non-volatile solute.
Final Answer : (2)

82) **Explaining :** Evaluate given assertion and reason.
Concept : This question is based on osmosis.
Solution :
When a solution is separated from the pure solvent by a semi-permeable membrane, the solvent molecules pass through it from pure solvent side to the solution side.
Diffusion of solute does not occurs from a region of high concentration solution to a region of low concentration of solution.
Final Answer : (3)

83) **Explaining :** A solution of acetone in ethanol :
Concept : This question is based on deviation from Raoult's law.
Solution : A solution of acetone in ethanol shows a positive deviation from Raoult's law.
Final Answer : (3)

84) **Explaining :** The mixture of two liquids boils without changing its composition is called:
Concept : This question is based on azeotropic mixtures.
Solution :

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Azeotropes are binary mixtures having the same composition in liquid and vapour phase and boils at a constant temperature.
Final Answer : (3)

85)

Explaining : Calculate elevation in boiling point for 2 molal aqueous solution of glucose:
Concept : This question is based on elevation in boiling point.
Solution : $\Delta T_b = K_b \times m = 0.5 \times 2 = 1°C$
Final Answer : (1)

86) **Explaining :** Cryoscopic constant of a liquid is:
Concept : This question is based on cryoscopic constant.
Solution :
$\Delta T_f = K_f \times m$
when $m = 1$
then, $K_f = \Delta T_f$
Final Answer : (2)

87) **Explaining :** What is ΔT_b for 0.2 molal aqueous solution of urea? (K_b of $H_2O = 0.52$)
Concept : This question is based on elevation in boiling point.
Solution :
$\Delta T_b = K_b m$
$= 0.52 \times 0.2$
$= 0.104$
Final Answer : (1)

88) **Explaining :** Evaluate given assertion and reason.
Concept : This question is based on molarity.
Solution : Molarity changes with change in temperature. It is not a colligative property.
Final Answer : (3)

89) **Explaining :** In which of the following cases blood cells will shrink:
Concept : This question is based on osmosis.
Solution :
When blood cells are placed in a solution containing more than 0.9% NaCl, the solution is hypertonic.
So water moves out of the cells by osmosis and they shrink.
Final Answer : (1)

90)

Explaining : Find out concentration terms independent of temperature.
Concept : This question is based on concentration terms.
Solution:
(a) Mass % → mass/mass → independent
(b) ppm → mass/mass → independent

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(c) Mole fraction → moles/moles → independent
(d) Molality → moles/wt. of solvent → independent
(e) Molarity → moles/volume → depends on temperature
Final Answer: (3)

BIOLOGY

91) **Solution : 1. Correct** – "A typical eukaryotic cell cycle is illustrated by human cells in culture. These cells divide once in approximately every 24 hours."
2. **Correct** – "Yeast, for example, can progress through the cell cycle in only about 90 minutes."
3. **Correct** – "The interphase represents the phase between two successive M phases."
4. **Correct** – "In the 24 hour average duration of cell cycle of a human cell, cell division proper lasts for only about an hour."
5. **Correct** – "During G1 phase the cell is metabolically active and continuously grows but does not replicate its DNA."
6. **Correct** – "During this time the amount of DNA per cell doubles. If the initial amount of DNA is denoted as 2C then it increases to 4C. However, there is no increase in the chromosome number."
Final Answer : (4)

92) **Explanation : Assertion** is **true** – DNA replication happens in **S phase**, which is part of **interphase**.
Reason is **false** – M phase is the **division phase** (mitosis), **not** the phase for DNA synthesis.
Final Answer : (3)

93) **Explanation :** Statement 2 is false: centriole duplication occurs during **S phase**, not G_2.
Statement 4 is false: mitosis in plants occurs in both haploid and diploid cells (NCERT).
Statements 1, 3, and 5 are correct per NCERT.
Final Answer : (1)

94) **Explanation :** These features (chromosome alignment at equator, kinetochores, nuclear envelope absent) are defining metaphase traits.
Final Answer : (2)

95) **Explanation :** Recombinase is the enzyme that facilitates the exchange of genetic material between homologous chromosomes during crossing over.
Final Answer : (3)

96) **Explanation:** Chromosomes align at the equator (metaphase plate), microtubules from opposite poles attach to kinetochores, and meiosis II results in four haploid daughter cells.
Final Answer : (1)

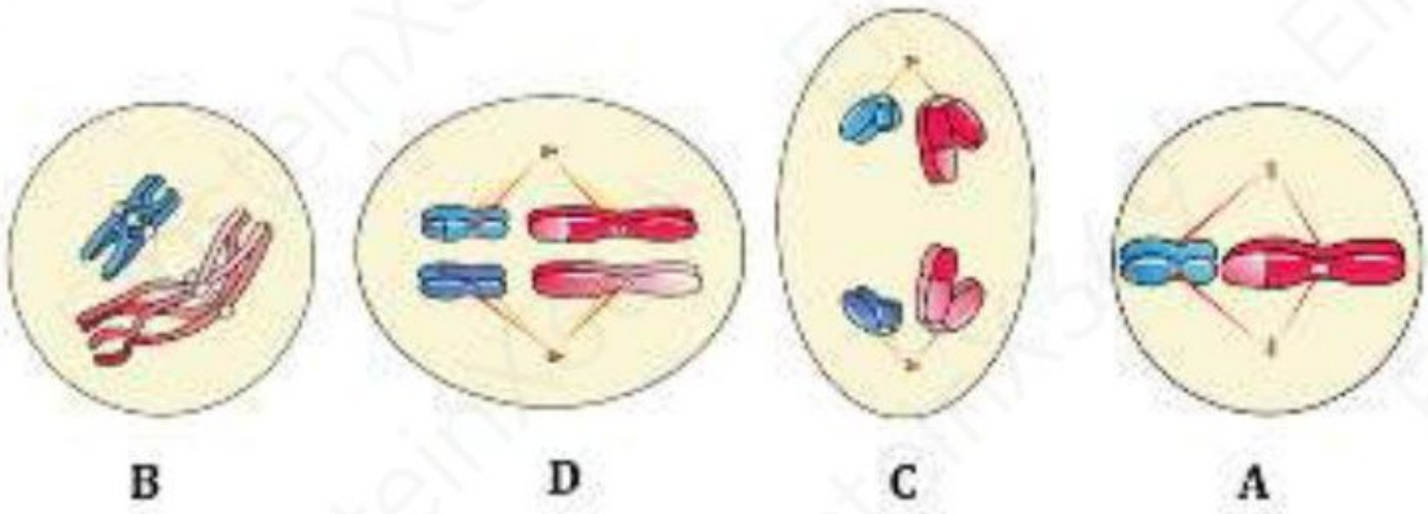
97) **Explanation:** Kinetochores are attachment points for spindle fibers. Without them, chromatids cannot be pulled apart.

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Final Answer : (2)

98) **Solution :** NCERT notes that leptotene is when chromosomes become visible; pachytene is when crossing over occurs.
Final Answer : (3)

99) **Solution :** B → D → C → A



Final Answer : (2)

100) **Solution :** Interkinesis is a brief interphase which sometimes occurs between meiosis I and meiosis II.
Final Answer : (1)

101) **Solution :** During telophase of mitosis, the nucleolous condense and reappear. The nuclear envelope is assembled around the chromatin mass and ER reform again thus forming two daughter nuclei.
Final Answer : (4)

102) **Solution :** All the given statements are correct.
Final Answer : (4)

103) **Solution :** The points of attachment between homologous chromosomes after their separation in diplotene appear X-shaped. They are called chiasmata.
Final Answer : (2)

104) **Solution :** (i) Chromatids separate but remains in the centre of the cell in metaphase.
(ii) Chromatids start moving towards opposite poles in anaphase.
(iii) Golgi complex and endoplasmic reticulum are still visible at the end of telophase.
Final Answer : (4)

105) **Solution :** All the statements regarding G_0 stage of the cell are correct. G_0 stage or resting phase is a period in the cell cycle in which cells exist in a quiescent state. G_0 phase is viewed as either an extended G_1 phase, where the cell is neither dividing nor preparing to divide, or a distinct quiescent stage that occurs outside of the cell cycle. Some (like cells, nerve and heart muscle cells), become quiescent when they reach maturity (i.e., when they are

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terminally differentiated) but continue to perform their main functions for the rest of the organism's life.
Final Answer : (4)

106) **Solution :** A – IV, B – III, C – II, D – I
Metaphase – Chromosomes are moved to spindle fibre.
Anaphase – Centromere splits and chromatids apart.
Zygotene – Pairing between homologous chromosomes takes place.
Pachytene – Crossing between homologous chromosomes occurs.
Final Answer : (3)

107) **Solution :** Cytokinesis is the division of cytoplasm, cytokinesis follows karyokinesis. Karyokinesis is the division of nucleus. The M phase starts with nuclear division, corresponding to the separation of daughter chromosomes and leads to division of cytoplasm.
Final Answer : (4)

108) **Solution :** Metaphase chromosome is made up of two sister chromatids which are held together by the centromere. Chromosomes are moved to spindle equator and gets aligned along metaphase plate through spindle fibres to both poles.
Final Answer : (3)

109) **Solution :** In some organisms karyokinesis is not followed by cytokinesis as a result of which multinucleate condition arises leading to the formation of syncytium (e.g., liquid endosperm in coconut).
Final Answer : (2)

110) **Solution :** In animal cell cytokinesis involves the contraction of the contractile ring of microfilament. Contractile ring of microfilaments develops peripherally below the cell membrane in the equatorial region.
Final Answer : (2)

111) **Solution :** Meiosis involves two sequential nuclear and cell division. It results in four daughter cells each with half the number of chromosomes of the parent cell, as in the production of gametes and plant spores. Meiosis begins with a parent cell that is diploid, meaning it has two copies of each chromosome. The parent cell undergoes one round of DNA replication followed by two separate cycles of nuclear division. The process results in four daughter cells that are haploid, which means they contain half the number of chromosomes of the diploid parent cell.
Final Answer : (1)

112) **Solution :** The correct sequence of the events in the meiosis is: III, II, I, IV.
III. Synapsis: Synapsis is the pairing of two homologous chromosomes that occurs during meiosis. It allows matching-up of homologous pairs prior to their segregation, and possible chromosomal crossover between them. It occurs in zygotene sage.
II. Crossing over: Crossing over is exchange of chromosomes segments between non-sister chromatids of homologous pair. Crossing over occurs during pachytene stage.

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I. Terminalization: Terminalization of chiasmata occurs in diakinesis stage of meiosis I. It is the disappearance of chiasmata of sliding towards the tips of the chromosomes due to tight condensation.
IV. Disjunction of genomes: The process of breakdown of homologous chromosomes is called disjunction. This process occurs in anaphase I.
Final Answer : (3)

113) **Solution :** Meiosis I is known as reductional division due to reduction in the number of chromosomes. Meiosis II is called equational division because of maintaining the same number of chromosomes.
Final Answer : (1)

114) **Solution :** Nucleolus and nuclear membrane always disappear during mitosis and meiosis. At the beginning of mitosis, the chromosomes condense, the nucleolus disappears, and the nuclear envelope breaks down, resulting in the release of most of the contents of the nucleus into the cytoplasm. At the end of mitosis, the process is reversed. In diakinesis stage of prophase I of meiosis I, nucleolus and nuclear membrane disappear and chromosomes set free in the cytoplasm.
Final Answer : (2)

115) **Solution :** The synaptonemal complex is formed at the region of synapsis. It is a protein structure that forms between homologous chromosomes (two pairs of sister chromatids) during meiosis and is thought to mediate chromosome pairing, synapsis, and recombination. This stage appears in zygotene stage of prophase I of meiosis I.
Final Answer : (4)

116) **Solution :** The proximal convoluted tubule (PCT) is lined by simple cuboidal epithelium with microvilli (brush border), which greatly increases the surface area for reabsorption. The alveoli and blood vessels are lined by simple squamous epithelium, not stratified epithelium.
Final Answer : (3)

117) **Solution :** Adhering junctions provide mechanical strength by cementing adjacent cells, whereas tight junctions prevent the leakage of substances across the intercellular space.
Final Answer : (3)

118) **Solution :** Statements A, B and D are correct. Statement C is incorrect because tendons connect muscles to bones, while ligaments connect bones to bones and are more elastic.
Final Answer : (1)

119) **Solution :** The diagram shows branched, striated fibres with intercalated discs, which are characteristic features of cardiac muscle.
Final Answer : (3)

120) **Solution :** Statements A, B and D are correct. Statement C is incorrect because diffusion and filtration are functions of simple squamous epithelium, whereas compound epithelium

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primarily provides protection.
Final Answer : (2)

121) **Solution :** Chondrocytes are the living cells of cartilage present within lacunae.
Final Answer : (3)

122) **Solution :** Statement C is incorrect because the ovaries of cockroach are not functionally connected to the kidneys.
Final Answer : (1)

123) **Solution :** The assertion is true—the abdomen of a cockroach has 10 segments. The reason is also true—the 10th abdominal segment bears a pair of anal cerci in both sexes. However, the presence of cerci does not explain why the abdomen has 10 segments. Hence, both A and R are true, but R is not the correct explanation of A.
Final Answer : (2)

124) **Solution :** Statement A is correct—the cockroach has a 13-chambered tubular heart. Statement B is incorrect because oxygen is transported through the tracheal system, not by haemolymph.
Final Answer : (3)

125) **Solution:** The immature stage of cockroach is called the **nymph**, which resembles the adult.
Final Answer : (4)

126) **Solution : (a)** represents a goblet cell, which is a unicellular exocrine gland. It secretes mucus directly onto the epithelial surface without forming a duct.
Final Answer : (1)

127) **Solution :** The sinus venosus receives blood through the venae cavae and opens into the right atrium. The conus arteriosus receives blood from the ventricle.
Final Answer : (2)

128) **Solution :** Statements A, B and D are correct. Statement C is incorrect because the urinogenital duct opens into the cloaca, not directly to the exterior.
Final Answer : (3)

129) **Solution :** Adult frogs are ureotelic because they excrete urea as their principal nitrogenous waste, so the reason correctly explains the assertion.
Final Answer : (3)

130) **Solution :** Frog skin is smooth, moist, glandular and without scales. It also participates in cutaneous respiration. It is not dry or heavily keratinised.

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Final Answer : (4)

131) **Solution :** The assertion is directly based on NCERT. However, prokaryotic cells lack membrane bound organelles, yet survive independently. Hence Reason is false.
Final Answer : (3)

132) **Solution :** NCERT states that cell wall provides structural support and prevents bursting or collapsing due to osmotic pressure.
Final Answer : (2)

133) **Solution :** Statement 2 is incorrect because centrosomes are absent in almost all plant cells.
Final Answer : (2)

134) **Solution :** Glycocalyx → Cell wall → Plasma membrane
(Glycocalyx may be capsule or slime layer.)
Final Answer : (3)

135) **Solution :** Chronological order:
• Antonie van Leeuwenhoek (live cell observed)
• Robert Brown (nucleus)
• Matthias Schleiden (plants made of cells)
• Rudolf Virchow (cells arise from pre-existing cells)
Final Answer : (1)

136) **Solution :** Singer and Nicolson's fluid mosaic model explains membrane fluidity. This fluidity permits membrane remodeling during endocytosis, secretion, cell division and formation of junctions.
Final Answer : (1)

137) **Solution :** Peripheral proteins remain on the membrane surface. Integral proteins are buried partially or completely.
Final Answer : (2)

138) **Solution :** Osmosis is the diffusion of water across a selectively permeable membrane from higher to lower water concentration.
Final Answer : (3)

139) **Solution :** Secondary wall is deposited inside the primary wall (towards the plasma membrane).
Final Answer : (2)

140) **Solution :** Only ER, Golgi, lysosomes and vacuoles constitute the endomembrane system. Mitochondria, chloroplasts and peroxisomes do not.

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Final Answer : (4)

141) **Solution :** Cristae are folds of the inner membrane that greatly increase its surface area for enzymes involved in aerobic respiration.
Final Answer : (1)

142) **Solution :** Cristae project towards the matrix, not into the intermembrane space.
Final Answer : (1)

143) **Solution :** Chromoplasts contain carotenoid pigments responsible for yellow, orange and red colours.
Final Answer : (3)

144) **Solution :** Chloroplast ribosomes are 70S, whereas cytoplasmic ribosomes of eukaryotes are 80S.
Final Answer : (2)

145) **Solution :** NCERT attributes mechanical support, cell shape maintenance and motility to the cytoskeleton.
Final Answer : (2)

146) **Solution :** 60S + 40S ≠ 100S because Svedberg unit is not additive. It depends on sedimentation behaviour, which is influenced by size, shape and density.
Final Answer : (1)

147) **Solution :** The axoneme of eukaryotic cilia and flagella exhibits the characteristic 9 + 2 arrangement.
Final Answer : (3)

148) **Solution :** Centrioles and basal bodies possess nine peripheral triplets. The 9 + 2 arrangement belongs to the axoneme, not the centriole.
Final Answer : (2)

149) **Solution :** Both statement-I & II are true.
Final Answer : (1)

150) **Solution :** Nuclear pores regulate the movement of RNA and proteins across the nuclear envelope.
Final Answer : (2)

151) **Solution :** mRNA synthesized inside the nucleus cannot leave through blocked nuclear pores, stopping protein synthesis first.

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Final Answer : (3)

152) **Solution :** Submetacentric chromosomes possess one shorter and one longer arm.
Final Answer : (2)

153) **Solution :** Kinetochores provide attachment sites for spindle fibres during chromosome movement.
Final Answer : (1)

154) **Solution :** Wheat grain cell.
Final Answer : (3)

155) **Solution :** Autophagy — Digesting worn off cell organelle by help of lysosomes.
Final Answer : (3)

156) **Solution :** RuBisCO is the most abundant protein in the whole of the biosphere. Proteins are linear chains of amino acids linked by peptide bonds. In a polysaccharide, the individual monosaccharides are linked by a glycosidic bond.
Final Answer : (1)

157) **Solution:** Formation of glycosidic bonds is a dehydration synthesis reaction (formation of bond with removal of water).
Final Answer : (2)

158) **Solution:** A – IV; B – I; C – II; D – III

Column-I		**Column-II**	
A	Cotton fibre	IV	Cellulose
B	Exoskeleton of cockroach	I	Chitin
C	Liver	II	Glycogen
D	Peeled potato	III	Starch

Final Answer : (2)

159) **Solution:** Cellulose does not show blue colour with iodine because it does not have helical structure while starch shows blue black colouration with iodine.
Final Answer : (3)

160) **Solution:** Palmitic acid – A saturated fatty acid with 18 carbon atoms. Adenylic acid - Adenosine with a phosphate molecule. Alanine amino acid – neutral amino acid.
Final Answer : (1)

161) **Solution :** Proteins – 10-15%; Nucleic acids – 5-7%; Carbohydrates – 3%; Lipids – 2%.

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Final Answer : (2)

162) **Solution:** Carbohydrates are primarily composed of carbon, hydrogen, and oxygen atoms.
Final Answer : (2)

163) **Solution:** Lipids are insoluble in water due to their non-polar nature.
Final Answer : (1)

164) **Solution :** The simplest carbohydrates are called monosaccharides, which include sugars like glucose and fructose. When two monosaccharides join, they form a disaccharide, such as sucrose.
Final Answer : (4)

165) **Solution: Malonate** structurally resembles **succinate** and inhibits **succinic dehydrogenase** by occupying its active site.
Final Answer : (2)

166) **Solution :** Monosaccharide – Sugar units
Starch – Energy storage in plants
Glycogen – Energy storage in animals
Phospholipid - Lecithin
Final Answer : (3)

167) **Solution:** Saturated fatty acids have all single bonds between carbon atoms.
Final Answer : (2)

168) **Solution:** Both statements are true.
Final Answer : (1)

169) **Solution:** Oils are liquid at room temperature.
Final Answer : (1)

170) **Solution:** All are true statements.
Final Answer : (3)

171) **Solution:** Living organisms have got both organic and inorganic elements and compounds in them. Oils have lower melting point and hence remain as liquid in winters.
Final Answer : (2)

172) **Solution:** Cell wall is absent in animals. Collagen is the most abundant protein in animal world.

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Final Answer : (4)

173) **Solution :** The **Reason is false**: Secondary metabolites are **not useless**; many have significant **ecological** and **economic** importance (e.g. antibiotics, rubber, spices, scents).
Final Answer : (3)

174) **Solution : Statement 2** – Incorrect: Unlike primary metabolites, **secondary metabolites are not essential** for growth or development.
Final Answer : (2)

175) **Solution:** Monosaccharide - Glucose
Disaccharide - Sucrose
Polysaccharide - Cellulose
Lipid - Palmitic acid
Nucleic acid - DNA and RNA
Final Answer : (3)

176) **Explanation:** RNA, a macromolecule, stays in the acid-insoluble fraction. Glucose and small metabolites are acid-soluble. Enzymes (proteins) also remain insoluble.
Final Answer : (2)

177) **Solution : 2** is incorrect because silicon is not a major component of biomolecules.
3 is incorrect because acid-soluble pool contains low molecular weight biomolecules.
4 is incorrect because trichloroacetic acid is used to precipitate macromolecules, not extract lipids.
Final Answer : (1)

178) **Solution: Oils** remain liquid due to the **presence of unsaturated fatty acids** (C=C bonds), which lower melting point. Fats typically have **saturated** fatty acids and are solid at room temp.
Final Answer : (3)

179) **Solution:** Proteins are heteropolymers, not homopolymers.
Final Answer : (2)

180) **Solution:** Quaternary structure involves spatial arrangement of multiple polypeptide chains.
Final Answer : (4)

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